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TEST BOOKLET SERIES

**Test Code:
2206****TEST BOOKLET
GENERAL STUDIES
TEST – 6****B****Time Allowed : Two Hours****Maximum Marks : 200****INSTRUCTIONS**

1. IMMEDIATELY AFTER THE COMMENCEMENT OF THE EXAMINATION, YOU SHOULD CHECK THAT THIS TEST BOOKLET *DOES NOT* HAVE ANY UNPRINTED OR TORN OR MISSING PAGES OR ITEMS, ETC. IF SO, GET IT REPLACED BY A COMPLETE TEST BOOKLET.
2. ENCODE CLEARLY THE TEST BOOKLET SERIES CODE **A, B, C** OR **D** AS THE CASE MAY BE IN THE APPROPRIATE PLACE IN THE ANSWER SHEET.
3. You have to enter your Roll Number on the Test Booklet in the Box provided alongside this line. *DO NOT* write *anything else* on the Test Booklet.
4. This Test Booklet contains 100 items (questions). Each question is printed both in **Hindi** and **English**. Each item comprises four responses (answers). You will select the response which you want to mark on the Answer Sheet. In case you feel that there is more than one correct response, mark the response which you consider the best. In any case, choose **ONLY ONE** response for each item.
5. You have to mark all your responses **ONLY** on the separate Answer Sheet provided. See directions in the Answer sheet.
6. All items carry equal marks.
7. Before you proceed to mark in the Answer Sheet the response to various items in the Test Booklet, you have to fill in some particulars in the Answer Sheet as per instructions sent to you with your admission certificate.
8. After you have completed filling in all your responses on the Answer Sheet and examination has concluded, you should hand over to the Invigilator *only the Answer Sheet*. You are permitted to take away with you the Test Booklet.
9. Sheets for rough work are appended in the Test Booklet at the end.
10. **Penalty for wrong answers:**
THERE WILL BE PENALTY FOR WRONG ANSWERS MARKED BY A CANDIDATE IN THE OBJECTIVE TYPE QUESTION PAPERS.
 - (i) There are four alternatives for the answer to every question. For each question for which a wrong answer has been given by the candidate, **One-Third** of the marks assigned to that question will be deducted as penalty.
 - (ii) If a candidate gives more than one answer, it will be treated as a **wrong answer** even if one of the given answers happens to be correct and there will be same penalty as above to that question.
 - (iii) If a question is left blank, i.e., no answer is given by the candidate, there will be no **penalty** for that question.

DO NOT OPEN THIS TEST BOOKLET UNTIL YOU ARE ASKED TO DO SO*Note:* Hindi version of the instructions is printed on the front cover of this Booklet.



1. With reference to the Kuiper Belt, which of the following statements is correct?
 - (a) It is an asteroid belt that lies between the orbits of Mars and Jupiter.
 - (b) It is the belt which works as a path for the comets revolving around the Sun.
 - (c) It is the region that lies beyond the eight planets of our solar system.
 - (d) It is the belt which holds the cluster of several galaxies.

2. Arrange the following stages involved in evolution of atmosphere and hydrosphere of earth in the chronological order:

1. Degassing
2. Loss of primordial atmosphere
3. Modification of composition of atmosphere through photosynthesis
4. Emergence of Oceans and other water bodies

Select the correct answer using the code given below:

- | | |
|-------------|-------------|
| (a) 1-2-4-3 | (b) 2-1-4-3 |
| (c) 4-2-1-3 | (d) 1-2-3-4 |

3. Consider the following statements about the Cluster Development Programme:

1. It has been announced to enhance the global competitiveness of the Indian horticulture sector.
2. The Department of Agriculture, Cooperation & Farmers Welfare has been designated as the Nodal Agency for implementation of the Programme.

Which of the statements given above is/are correct?

- | | |
|------------------|---------------------|
| (a) 1 only | (b) 2 only |
| (c) Both 1 and 2 | (d) Neither 1 nor 2 |

4. Which of the following is/are characteristic feature of Chinook winds?

1. They are highly moist and dense winds experienced on the windward side of mountains.
2. They can cause significant temperature spikes in the area they flow through.

Select the correct answer using the code given below:

- | | |
|------------------|---------------------|
| (a) 1 only | (b) 2 only |
| (c) Both 1 and 2 | (d) Neither 1 nor 2 |

5. With reference to the Atlantic Meridional Overturning Circulation (AMOC), consider the following statements:

1. AMOC carries cold water from the North Atlantic towards the tropics.
2. AMOC aids in distributing heat and energy around the earth.

Which of the statements given above is/are correct?

- | | |
|------------------|---------------------|
| (a) 1 only | (b) 2 only |
| (c) Both 1 and 2 | (d) Neither 1 nor 2 |

6. With reference to mCessation Programme, which of the following statements is/are correct?

1. It is an initiative of the National Tobacco Control Programme and the Union Ministry of Health and Family Welfare, with support from the WHO and the International Telecommunication Union.
2. It is an initiative for utilising mobile technology for tobacco cessation.
3. It uses two-way messaging with the individual seeking to quit tobacco use and programme specialists providing them dynamic support.

Select the correct answer using the code given below:

- | | |
|------------|------------------|
| (a) 1 only | (b) 1 and 2 only |
| (c) 3 only | (d) 1, 2 and 3 |

7. Which of the following factors cause variation in insolation?

1. The rotation of earth.
2. The angle of inclination of the sun's rays.
3. The length of the day.

Select the correct answer using the code given below:

- | | |
|------------------|------------------|
| (a) 1 and 2 only | (b) 2 and 3 only |
| (c) 1 and 3 only | (d) 1, 2 and 3 |





8. What is the correct sequence of occurrence of the following cities in Central Asia as one proceeds from North to South?

- | | |
|---------------|-------------|
| 1. Dushanbe | 2. Tashkent |
| 3. Nur-Sultan | 4. Bishkek |

Select the correct answer using the code given below.

- | | |
|-------------|-------------|
| (a) 4-2-1-3 | (b) 3-2-4-1 |
| (c) 3-4-2-1 | (d) 4-3-2-1 |

9. With reference to the Shanghai Cooperation Organisation (SCO), consider the following statements:

1. The creation of the SCO was announced on 15 June 2001 in Beijing (China).
2. The SCO Charter entered into force on 19 September 2003.
3. SCO has two permanent bodies namely the SCO Secretariat and the Executive Committee of the Regional Anti-Terrorist Structure (RATS).
4. As of November 2021, the SCO comprises ten member states.

Which of the statements given above are correct?

- | | |
|---------------------|---------------------|
| (a) 1, 2 and 3 only | (b) 1, 3 and 4 only |
| (c) 2 and 3 only | (d) 1, 2, 3 and 4 |

10. The Great Barrier Reef is located at which of the following?

- (a) Off the coast of Solomon Islands
- (b) Off the coast of Queensland in Northeastern Australia
- (c) At Timor Sea in Southern Australia
- (d) South of Kimberley Plateau

11. Consider the following statements:

1. Volcanoes are always found over the subduction zone near plate boundaries.
2. San Andreas Fault is located between Pacific and the North American plate.

Which of the statements given above is/are correct?

- | | |
|------------------|---------------------|
| (a) 1 only | (b) 2 only |
| (c) Both 1 and 2 | (d) Neither 1 nor 2 |

12. Consider the following pairs:

Space Mission	Space Agency
1. DAVINCI+	: Japan Aerospace Exploration Agency
2. EnVision	: European Space Agency
3. VERITAS	: National Aeronautics and Space Administration (NASA)

Which of the pairs given above is/are correctly matched?

- | | |
|------------------|------------------|
| (a) 1 only | (b) 1 and 3 only |
| (c) 2 and 3 only | (d) 1, 2 and 3 |

13. With reference to the concept of Inversion of Temperature, consider the following statements:

1. Normal lapse rate is the degree of increase in temperature with decrease in elevation.
2. Air drainage is the concept where during night cold air produced at the hills and mountains flows under the influence of gravity and moves down the slope to pile up valley bottoms with warm air above.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

14. Arrange the following countries from west to east around the Persian Gulf?

1. Kuwait
2. United Arab Emirates
3. Bahrain
4. Oman

Select the correct answer using the code given below:

- | |
|-------------|
| (a) 2-3-4-1 |
| (b) 1-2-3-4 |
| (c) 1-3-2-4 |
| (d) 2-3-1-4 |



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15. The 'World Employment and Social Outlook: Trends 2021' report is published by the
- World Bank
 - United Nations Conference on Trade and Investment
 - World Economic Forum
 - International Labour Organisation
16. "Actions to protect, sustainably manage, and restore natural or modified ecosystems, that address societal challenges effectively and adaptively, simultaneously providing human well-being and biodiversity benefits" define
- Biodiversity Services
 - Nature-based Solutions
 - Sustainable Development
 - Ecosystem Services approach
17. With reference to the Mass Movements in geography, consider the following statements:
- Mass movements are aided by gravity and no geomorphic agent like running water or wind participates in this.
 - Weathering is a pre-requisite for mass movements.
- Which of the statements given above is/are correct?
- 1 only
 - 2 only
 - Both 1 and 2
 - Neither 1 nor 2
18. Consider the following statements about Sustainable Development Goals (SDG) India Index:
- The index was first launched in 2018.
 - It evaluates the progress of states and UTs on parameters such as health, education, gender, economic growth, institutions, climate change and environment.
 - The index is developed in collaboration with the United Nations in India.
 - In the SDG India Index 2020-21, Himachal Pradesh was the top performer while Bihar was ranked last.

Which of the statements given above are correct?

- 1, 2 and 3 only
- 3 and 4 only
- 2, 3 and 4 only
- 1, 2, 3 and 4

19. Consider the following statements:

- River terraces are formed due to the lateral erosion by the stream.
- Meander is not a landform but is only a type of channel pattern.

Which of the statements given above is/are correct?

- 1 only
- 2 only
- Both 1 and 2
- Neither 1 nor 2

20. In South-East Asia, which among the following countries is landlocked?

- Laos
- Vietnam
- Cambodia
- Thailand

21. Consider the following statements:

- In 1970, the UN General Assembly designated 5 June as World Environment Day (WED).
- The first celebration of the WED took place in 1974.
- Pakistan was the global host country for World Environment Day 2021.
- The theme for World Environment Day 2021 was 'Only One Earth'.

Which of the statements given above are correct?

- 1 and 3 only
- 1 and 4 only
- 2 and 3 only
- 3 and 4 only

22. Consider the following statements:

- Goldilocks zone is the area around a star where it is not too hot and not too cold for liquid water to exist on the surface of surrounding planets.
- NASA's DART Mission is envisaged to trace the exo-planets that lie in Goldilocks zone outside the solar system.

Which of the statements given above is/are correct?

- 1 only
- 2 only
- Both 1 and 2
- Neither 1 nor 2



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23. Soil formation is mainly related to:

1. Surface Relief
2. Climate
3. Natural vegetation
4. Parent rock material

Select the correct answer using the code given below:

- (a) 1, 2 and 3 only (b) 2, 3 and 4 only
(c) 1, 3 and 4 only (d) 1, 2, 3 and 4

24. Consider the following statements:

1. The SAGE initiative and the SAGE portal has been launched by the Ministry of Health & Family Welfare.
2. The SAGE programme and SAGE portal have been launched with a view to help such persons who are interested in entrepreneurship in the field of providing services for elderly care.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

25. With reference to the Amazon rainforest, consider the following statements:

1. More than 80 percent of the rainforest is located within Brazil.
2. Amazon rainforest covers nine countries in South America.

Which of the statements given above is/are correct?

- (a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

26. Consider the following statements about the Madden-Julian Oscillation (MJO):

1. It is a stationary system of warm water observed over the Indian Ocean.
2. It occurs at intervals of 3-7 years.
3. This phenomenon always leads to higher monsoon rains in India.

Which of the statements given above are correct?

- (a) 1 and 3 only (b) 2 and 3 only
(c) 1, 2 and 3 (d) None of the above

27. The system of production, distribution and consumption of goods and services aimed at using the purchasing potential of older and ageing people and satisfying their consumption, living and health needs is called:

- (a) Golden economy
(b) Iron economy
(c) Silver economy
(d) None of the above

28. Despite being in the cool temperate region, Western Europe does not experience extreme climate because:

1. The western margins of Europe are under the influence of easterlies throughout the year.
2. The North Atlantic drift moderates the overall temperature.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

29. With reference to Cosmic Infrared Background Experiment-2 (CIBER-2), which of the following statements is/are correct?

1. The aim of this mission is to count the number of stars that exist in the Milky Way.
2. For counting the stars, CIBER-2 will detect the extragalactic background light.
3. This is the first time that such a mission has been undertaken.

Select the correct answer using the code given below:

- (a) 1 and 2 only
(b) 2 only
(c) 3 only
(d) 1, 2 and 3



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30. With reference to the Vaigai river, which of the following statements is **not** correct?

- Vaigai basin is bounded by the Varusanadu Hills, the Andipatti Hills, the Cardamom Hills and the Palani Hills on the West and by the Palk Strait and Palk Bay on the East.
- Vaigai River rises near Kottaimalai in the Madurai district.
- It traverses a path of about 258 Km.
- The ib, the Ong and the Tel are some of the tributaries of the Vaigai river.

31. Consider the following statements:

- Dry air has less capacity to hold water vapour than humid air.
- Absolute humidity is the essential characteristic of weather forecasts.
- Higher the amount of water vapour, higher is the absolute humidity.

Which of the statements given above is/are correct?

- 1 and 2 only
- 2 and 3 only
- 3 only
- 1, 2 and 3

32. With reference to the 'Cold Deserts', consider the following statements:

- Cold Deserts witness prolonged and dry Summer.
- These deserts don't remain cold the entire year like polar deserts.
- These are found at high altitude and in rain shadow regions of temperate regions.

Which of the statements given above is/are correct?

- 1 only
- 1 and 2 only
- 2 and 3 only
- 1, 2 and 3

33. Consider the following statements about the Organisation for the Prohibition of Chemical Weapons (OPCW):

- It is an international organization established under the Chemical Weapons Convention in 1989.
- It has 193 Member States.
- The OPCW was awarded the Nobel Peace Prize in 2017.

4. It is headquartered in Hague, Netherlands.

Which of the statements given above are correct?

- 1, 2 and 3 only
- 1, 3 and 4 only
- 2 and 4 only
- 1, 2, 3 and 4

34. Consider the following statements about seafloor hydrothermal vents:

- Metal sulfides and several noble metals can be recovered from such systems.
- These vent systems are amongst some of the most productive ecosystems on earth.

Which of the statements given above is/are correct?

- 1 only
- 2 only
- Both 1 and 2
- Neither 1 nor 2

35. Consider the following statements:

- The Equator roughly passes through the middle of Africa dividing it into two nearly equal parts.
- Africa is the only continent through which the Tropic of Cancer, the Equator and the Tropic of Capricorn pass.

Which of the statements given above is/are correct?

- 1 only
- 2 only
- Both 1 and 2
- Neither 1 nor 2

36. With reference to the Performance Grading Index (PGI) 2019-20, consider the following statements:

- PGI for States and Union Territories was first published in 2019 with the reference year 2017-18.
- PGI is a tool to provide insights on the status of college education in States and Union Territories.
- PGI 2019-20 has been released by the Department of Higher Education.

Which of the statements given above is/are **not** correct?

- 1 and 2 only
- 2 and 3 only
- 3 only
- 1, 2 and 3



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37. With reference to volcanic landforms, consider the following statements:

1. Volcanic rocks are formed by cooling of the lava at the crust.
2. Plutonic rocks are formed by cooling of the lava at the surface.

Which of the statements given above is/are **not** correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

38. With reference to the Coral Triangle, consider the following statements:

1. The Coral Triangle is a geographical term for a roughly triangular area of the tropical marine waters of Indonesia, Malaysia, Papua New Guinea, Philippines, Solomon Islands and Timor-Leste.
2. It has all the coral species of the world.
3. All 7-turtle species are found here.

Which of the statements given above is/are correct?

- (a) 1 only
(b) 1 and 2 only
(c) 1 and 3 only
(d) 1, 2 and 3

39. Consider the following statements:

1. Recently, Turkey's Sea of Marmara witnessed the largest outbreak of 'sea snail'.
2. 'Sea snail' is marine mucilage that is formed when algae are overloaded with nutrients as a result of water pollution combined with the effects of climate change.
3. The Sea of Marmara connects the Aegean Sea to the Caspian sea.

Which of the statements given above is/are correct?

- (a) 1 and 2 only (b) 1 and 3 only
(c) 2 only (d) 1, 2 and 3

40. With reference to Karst landforms, consider the following statements:

1. Karst landforms are developed in dry climatic regions.
2. Dissolving process plays a key role in formation of these landforms.
3. Caves and Uvala are erosional Karst landforms.

Which of the statements given above are correct?

- (a) 1 and 2 only
(b) 1 and 3 only
(c) 2 and 3 only
(d) 1, 2 and 3

41. With reference to seismic waves, consider the following statements:

1. P-waves can travel only through solid medium whereas S-waves can travel through solids as well as liquids.
2. S-waves are longitudinal waves.
3. P-waves are the first waves to be detected on seismographs.

Which of the statements given above is/are **not** correct?

- (a) 1 and 3 only
(b) 3 only
(c) 1 and 2 only
(d) 2 and 3 only

42. Which among the following has launched a new global database named "I-Familia" to identify missing persons through family DNA and help the police solve cold cases in member countries?

- (a) United Nations Office on Drugs and Crime
(b) European Police Office
(c) Interpol
(d) Police Community of the Americas



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43. With reference to YUVA Scheme, consider the following statements:
1. The scheme aims to train 100 aspiring writers below 40 years, who are ready to express themselves and project India and its culture and literature globally.
 2. A consolidated scholarship of Rs 50,000 per month for a period of six months per author will be paid under the mentorship scheme.
- Which of the statements given above is/are correct?
- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2
44. Mid-Oceanic Ridges primarily occur along which of the following?
- (a) Divergent plate boundaries
(b) Convergent plate boundaries
(c) Transform plate boundaries
(d) Abyssal Plains
45. Consider the following pairs:
- | National Park | State |
|-------------------|---------------------|
| 1. Dihing Patkai | – Arunachal Pradesh |
| 2. Raimona | – Nagaland |
| 3. Orang | – Assam |
| 4. Dibru-Saikhowa | – Mizoram |
- Which of the pairs given above are **not** correctly matched?
- (a) 1, 2 and 4 only (b) 2, 3 and 4 only
(c) 2 and 4 only (d) 1, 2, 3 and 4
46. Consider the following statements:
1. White dwarf stars are formed when the hydrogen used as nuclear fuel by the star is exhausted.
 2. All the white dwarf stars cool and transform into black dwarfs over a period of time.
- Which of the statements given above is/are correct?
- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2
47. Which of the following countries share its border with the Baltic Sea?
1. Denmark
 2. Norway
 3. Russia
 4. Finland
 5. Kazakhstan
- Select the correct answer using the code given below:
- (a) 1, 3 and 5 only (b) 2, 3 and 4 only
(c) 1, 3 and 4 only (d) 1, 2, 3 and 4 only
48. With reference to International Yoga Day (IYD), consider the following statements:
1. In 2021, the 7th annual International Yoga Day was celebrated.
 2. The United Nations theme for the year 2021 was “Yoga for well-being”.
 3. The UN proclaimed June 21 as International Day of Yoga by passing a resolution on December 11, 2014.
- Which of the statements given above is/are correct?
- (a) 1 only (b) 2 and 3 only
(c) 3 only (d) 1, 2 and 3
49. Consider the following statements:
1. Precipitation around the globe is more over land than the precipitation over oceans.
 2. The amount of water held as soil moisture is more than the quantity of water contained by freshwater lakes.
 3. The amount of water contained by an air parcel depends on the air temperature.
- Which of the statements given above is/are correct?
- (a) 1 and 2 only (b) 2 only
(c) 3 only (d) 2 and 3 only
50. What is the correct sequence of occurrence of the following lakes in Africa as one proceeds from north to south?
1. Lake Malawi
 2. Lake Tanganyika
 3. Lake Turkana
 4. Lake Tana





Select the correct answer using the code given below:

- (a) 1-2-3-4 (b) 4-3-2-1
(c) 2-3-4-1 (d) 1-3-2-4

51. Consider the following statements about Black Softshell Turtle:

1. It is listed as 'extinct in the wild' category of the International Union for Conservation of Nature red list.
2. It is found in ponds of temples in northeastern India and Bangladesh.
3. It enjoys legal protection under the Indian Wildlife (Protection) Act of 1972.

Which of the statements given above is/are correct?

- (a) 1 and 2 only (b) 2 only
(c) 1 and 3 only (d) 2 and 3 only

52. The Tropic of Capricorn passes through which of the following countries?

1. Madagascar
2. Indonesia
3. Brazil
4. Australia

Select the correct answer using the code given below:

- (a) 2 and 4 only (b) 2 and 3 only
(c) 1, 3 and 4 only (d) 4 only

53. Which of the following statements is/are correct?

1. In middle and high latitudes, the west coasts of continents are bordered by cold currents.
2. West coasts of the continents in tropical and sub-tropical latitudes (except close to the equator) are bordered by warm waters.
3. The mixing of warm and cold current results in best fishing grounds of the world.

Select the correct answer using the code given below:

- (a) 1 only
(b) 3 only
(c) 1 and 2 only
(d) None of the above.

54. Which one of the following terms that were widely heard in news recently is used for a type of research that modifies a biological agent so that it confers new or enhanced activity to that agent?

- (a) Analysis of Covariance
(b) Gain-of-Function Research
(c) Adjusted R-Squared
(d) Conditional Probability Models

55. Coriolis force plays a vital role in both atmospheric circulation and oceanic circulation. Consider the following statements about Coriolis force:

1. It is generated by rotation of the earth.
2. It is maximum at the poles and is absent at the equator.

Which of the statements given above is/are correct?

- (a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

56. Match the following pairs:

LIST-I

- A. Malacca Strait
B. Hormuz Strait
C. Gibraltar Strait
D. Dover Strait

LIST-II

1. Andaman Sea and South China Sea
2. Mediterranean Sea and Atlantic Sea
3. Persian Gulf and Gulf of Oman
4. English Channel and Northern Sea

Select the correct answer using the code given below:

- | | A | B | C | D |
|-----|---|---|---|---|
| (a) | 1 | 2 | 3 | 4 |
| (b) | 3 | 2 | 4 | 1 |
| (c) | 1 | 3 | 2 | 4 |
| (d) | 3 | 1 | 4 | 2 |



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57. With reference to the Biotech-KISAN scheme, consider the following statements:

1. It is a scheme of the Ministry of Agriculture & Farmers Welfare.
2. It is a state-specific program, following a hub-and-spoke model and stimulates entrepreneurship and innovation in farmers and empowers women farmers.
3. It aims to understand the problems of water, soil, seed and market faced by the farmers and provide simple solutions to them.

Which of the statements given above is/are correct?

- (a) 1 and 2 only (b) 2 and 3 only
(c) 3 only (d) 1, 2 and 3

58. The rate of evaporation increases with increase in:

1. Humidity
2. Temperature
3. Wind Speed
4. Surface Area

Select the correct answer using the code given below:

- (a) 1, 2 and 3 only (b) 2, 3 and 4 only
(c) 1, 3 and 4 only (d) 1, 2, 3 and 4

59. They are found in warm and humid areas with an annual precipitation of over 200 cm and mean annual temperature above 22°C. Which of the following forest is this:

- (a) Evergreen forests
(b) Semi-evergreen forests
(c) Moist Deciduous forests
(d) Littoral and Swamp forests

60. With reference to Herbicide Tolerant (HT) Bt Cotton, which of the following statements is/are correct?

1. The HTBt cotton is approved by the Genetic Engineering Appraisal Committee.
2. The HTBt cotton variant adds another layer of modification, making the plant resistant to the herbicide glyphosate.
3. Bt cotton and Bt mustard are the only transgenic crops that are approved by the Centre for commercial cultivation in India.

Select the correct answer using the code given below:

- (a) 1 only (b) 1 and 3 only
(c) 2 only (d) 1, 2 and 3

61. Which of the following countries do **not** share their border with the Black Sea?

1. Georgia
2. Serbia
3. Romania
4. Moldova
5. Ukraine

Select the correct answer using the code given below:

- (a) 1, 2 and 5 only (b) 2 and 4 only
(c) 2, 3 and 4 only (d) 1, 3, 4 and 5 only

62. Basket-of-eggs topography is related to

- (a) Drumlins (b) Eskers
(c) Cirques (d) Moraines

63. With reference to the 'UN Global Compact', consider the following statements:

1. It is a long-term agenda, related to sustainable development goals.
2. It is binding to the global businesses regarding their responsible policies.
3. The CEO Water Mandate is an initiative of the UN Global Compact.

Which of the statements given above is/are correct?

- (a) 1 and 2 only (b) 3 only
(c) 1 and 3 only (d) 2 and 3 only

64. With reference to Mediterranean climate, consider the following statements:

1. It is characterised by hot and wet summer and dry and mild winter.
2. The Mediterranean climate owes its origin to the seasonal shifting of pressure belts due to northward and southward migration of the sun.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2



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65. With reference to the geostrophic wind, consider the following statements:

1. The geostrophic wind blows parallel to the isobar.
2. These winds are free from frictional effect of the surface and are controlled by the pressure gradient and the Coriolis force only.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

66. With reference to the 'overture supersonic jet', which was recently in the news, consider the following statement:

1. It will be the fastest supersonic jet ever used.
2. It will ensure net-zero carbon emission as well as zero overland noise.
3. It will be highly cost-effective compared to other supersonic jets.

Which of the statements given above is/are correct?

- (a) 1 only (b) 1 and 2 only
(c) 3 only (d) 1, 2 and 3

67. Consider the following statements regarding Heat Budget:

1. In Tropical region, insolation received is higher than the high latitudes.
2. Equator receives more solar radiation than the tropics.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

68. Consider the following statements about Savanna type climate:

1. It is located between the tropic and temperate regions.
2. It is characterized by high rainfall throughout the year.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

69. With reference to 'Gravitational Lensing', which of the following statements is/are correct?

1. It occurs when a huge amount of matter, such as a massive galaxy, cluster of galaxies or a black hole, creates a gravitational field that distorts and magnifies the light from objects behind it.
2. It is based on Einstein's theory of general relativity.
3. It will be used by NASA's James Webb Space Telescope (JWST) for astronomical observations.

Select the correct answer using the code given below:

- (a) 1 only (b) 2 only
(c) 1 and 2 only (d) 1, 2 and 3

70. With reference to Operation Pangea, which of the following statements is/are correct?

1. It is an international initiative to control the online marketing of automatic firearms.
2. It is coordinated globally by Interpol.

Select the correct answer using the code given below:

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

71. The Generalised Scheme of Preference Plus (GSP+), which was recently in the news is related to

1. Providing tariff advantage to vulnerable low and lower-middle-income countries of Africa
2. Encourage sustainable development and good governance

Select the correct answer using the code given below:

- (a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2



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72. With reference to the World Competitiveness Index 2021, India's performance has improved in which of the following indices compared to the previous year?

- (a) Economic performance
- (b) Government efficiency
- (c) Business efficiency
- (d) Infrastructure

73. With reference to Hurricanes, consider the following statements:

1. Hurricanes are a type of tropical cyclone that occurs in the northeastern Pacific Ocean only.
2. They are formed over the warm ocean waters near the equator.
3. They typically form between 5 to 15 degrees latitude north and south of the equator.
4. They are usually not so strong and rarely cause heavy rains, deadly waves and fierce winds.

Which of the statements given above are correct?

- (a) 1 and 2 only
- (b) 2 and 4 only
- (c) 2 and 3 only
- (d) None of the above

74. Consider the following statements about the Black Holes:

1. The term 'Black Hole' was coined by Albert Einstein.
2. It is a point in space where matter is so compressed that it creates a gravity field from which even light cannot escape.
3. Black-Holes were theorized by John Archibald Wheeler.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 1 and 2 only
- (c) 2 only
- (d) 2 and 3 only

75. With reference to the 'Doppler Weather Radar', which was recently seen in the news, consider the following statements:

1. Doppler Weather Radar uses parabolic dish antennas to improve precision in long-range weather forecasting.

2. Doppler Weather Radar with a larger wavelength is more sensitive and can detect smaller particles.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

76. With reference to the El Niño-Southern Oscillation (ENSO), consider the following statements:

1. It causes both the increase and decrease of the temperature of the tropical Pacific Ocean.
2. During ENSO neutral conditions, surface trade winds blow eastward across the equatorial Pacific Ocean.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

77. Which of the following statements is/are correct about the Ramgarh Vishdhari Wildlife Sanctuary of Rajasthan?

1. It became the 52nd Tiger Reserve of India.
2. It became the 5th Tiger Reserve of Rajasthan.
3. It is located in the western part of Rajasthan.

Select the correct answer using the code given below:

- (a) 1 only
- (b) 2 only
- (c) 1 and 3 only
- (d) 2 and 3 only

78. Recently, under the Pygmy Hog Conservation Programme, eight captive-bred Pygmy Hogs were released in the Manas National Park of Assam. Which of the following statements is/are correct about the Pygmy Hog?

1. It is the world's smallest wild pig.
2. It is considered an indicator species.
3. It is adapted more to the wet grassland.

Select the correct answer using the code given below:

- (a) 1 and 2 only
- (b) 2 and 3 only
- (c) 3 only
- (d) 1, 2 and 3



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79. With reference to ocean acidification, consider the following statements:

1. In ocean acidification, carbon dioxide reacts with seawater and changes the pH of the ocean.
2. It helps in the growth of coral reefs.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

80. With reference to Tsunamis, consider the following statements:

1. They have high frequency over deep waters.
2. They lose very little energy while propagating.
3. Most of the Tsunamis are caused by frequent volcanic eruptions in the Pacific Ring of Fire.

Which of the statements given above is/are correct?

- (a) 1 and 2 only (b) 2 only
(c) 2 and 3 only (d) 1, 2 and 3

81. With reference to the 'Antarctic Treaty,' which of the following statements is/are correct?

1. It came into force in 1959.
2. India is a member of this treaty.
3. It aims to demilitarize the Antarctic region.

Select the correct answer using the code given below:

- (a) 1 and 2 only (b) 2 only
(c) 1 and 3 only (d) 2 and 3 only

82. With reference to the sea floor spreading, consider the following statements:

1. Sea floor spreading occurs at transform plate boundaries.
2. The age, density, and thickness of oceanic crust increases with distance from the mid-ocean ridge.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

83. Arrange the following mountains of Africa from north to south:

1. Atlas
2. Cameroon
3. Drakensberg

Select the correct answer using the code given below

- (a) 1-2-3 (b) 2-3-1
(c) 3-1-2 (d) 2-1-3

84. With reference to the National Maritime Heritage Complex, consider the following statements:

1. It would be developed with the idea of creating education with an entertainment approach.
2. It will provide a platform for the states to showcase their rich maritime heritage.
3. It will be developed in the Cochin region of Kerala.

Which of the statements given above is/are correct?

- (a) 1 and 3 only (b) 1 and 2 only
(c) 3 only (d) 1, 2 and 3

85. With reference to 'Polar Vortex', consider the following statements:

1. It is a large area of low pressure and cold air surrounding both the Earth's poles.
2. This phenomenon weakens in summer and strengthens in winter.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

86. With reference to Dead Zones in oceans, consider the following statements:

1. These are areas in the sea where lack of oxygen makes it difficult for organisms to survive.
2. Pollution through nutrient overload (Eutrophication) by humans is one of the causes of these zones.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2



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87. With reference to the recently developed, first Genetically Modified (GM) rubber, consider the following statements:

1. It is developed by the Indian Agricultural Research Institute (IARI).
2. It can withstand severe cold and drought conditions.
3. It will meet the long maturity period concern regarding rubber production in India.

Which of the statements given above is/are correct?

- (a) 1 and 2 only (b) 2 only
(c) 1 and 3 only (d) 2 and 3 only

88. Consider the following statements:

1. Deserts like Atacama of South America and Kalahari of Africa are part of the horse latitudes.
2. The top of the troposphere is higher in summer than in winter.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

89. With reference to the temperature of ocean water, consider the following statements:

1. The highest surface temperature of ocean water is recorded at slightly south of the equator.
2. The layer of ocean which experiences steep fall in temperature is known as thermocline.
3. In the arctic circle, there is only one layer of cold water which extends from surface to deep ocean floor.

Which of the statements given above are correct?

- (a) 1 and 2 only (b) 2 and 3 only
(c) 1, 2 and 3 (d) 1 and 3 only

90. With reference to the Integrated Power Development Scheme (IPDS), consider the following statements:

1. It is a dedicated power solutions scheme, only for the urban areas.
2. IT enablement is one of the focused components of this scheme.

3. Financial assistance under this scheme is available only for the public sector power distribution companies.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) 1 and 2 only (d) 1, 2 and 3

91. With reference to the 'Land for Life Award 2021', which among the following statements is **not** correct?

- (a) Land for Life Award is a prestigious award to recognize excellence and innovation in efforts towards land in balance.
- (b) It is given by Greenpeace international.
- (c) Award for the year 2021 was given for the environment conservation concept, 'Familial Forestry'.
- (d) Award for the year 2021 was received by an Indian climate activist.

92. With reference to Coral Reef, consider the following statements:

1. Red sea does not have any form of coral reefs.
2. El Nino is posing threat to coral diversity of Great Barrier Reef.
3. Area of the cold water (deep) coral is more than the occupied area of shallow water coral reefs around the world.

Which of the statements given above are correct?

- (a) 1 and 2 only (b) 1 and 3 only
(c) 2 and 3 only (d) 1, 2 and 3

93. Consider the following statements about the 'Gujarat International Maritime Arbitration Centre' (GIMAC) which was recently in the news:

1. It will resolve commercial and financial conflicts between entities having operations in India.
2. It is India's first arbitration centre related to maritime and shipping sector disputes.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2



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94. Consider the following statements regarding the dew formation:

1. Cold and long nights support dew formation.
2. Dew forms only when dew point is above the freezing point.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

95. Consider the following statements:

1. Fossil fuel can be found in the Continental shelf.
2. Seamount is a mountain of volcanic origin, which rises from the seafloor and does not reach the surface of the ocean.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

96. Consider the following statements regarding the Minimum Support Price (MSP):

1. While recommending MSP, the price of crops in Domestic as well as International Market is taken into consideration.
2. MSP is announced only for food crops.
3. The MSP recommended by the Commission for Agricultural Costs and Prices is final and binding on the government.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 and 3 only
(c) 1 and 2 only (d) 3 only

97. Consider the following statements regarding Ocean Currents:

1. Deep ocean currents are driven by density and temperature gradients.
2. Vertical movement of ocean currents is observed in upwelling and downwelling.
3. Surface currents make up around 70% of all water in the ocean.

Which of the above statements is/are correct?

- (a) 1 only
(b) 3 only
(c) 1 and 2 only
(d) 1, 2 and 3

98. Consider the following statements about Kelp forests:

1. Their distribution range is limited to the polar regions of the arctic.
2. These are underwater forests found in deep oceanic waters.
3. Climate change has been affecting kelp forests adversely.

Which of the statements given above are **not** correct?

- (a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

99. With reference to the New Atlantic Charter, consider the following statements:

1. It was signed by the members of G7.
2. It is the collective security pact.
3. It is a successor pact to the North Atlantic Treaty Organization.

Which of the statements given above is/are **not** correct?

- (a) 3 only (b) 1 and 2 only
(c) 2 and 3 only (d) 1, 2 and 3

100. Consider the following statements:

1. Spratly islands are uninhabited islands in the South China Sea.
2. China and Vietnam are the only countries involved in the Spratly island dispute.
3. Paracel islands are located north of the Spratly islands.

Which of the statements given above are correct?

- (a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3



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Space for Rough Work
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Space for Rough Work
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Test Code:
2206

(English Version)

ANSWER & EXPLANATION
GENERAL STUDIES
TEST – 06
B
ANSWERS

Q. No.	Answers	Q. No.	Answers	Q. No.	Answers	Q. No.	Answers
1	(c)	26	(d)	51	(b)	76	(a)
2	(b)	27	(c)	52	(c)	77	(a)
3	(a)	28	(b)	53	(b)	78	(d)
4	(b)	29	(b)	54	(b)	79	(a)
5	(c)	30	(d)	55	(c)	80	(b)
6	(d)	31	(c)	56	(c)	81	(d)
7	(d)	32	(c)	57	(c)	82	(b)
8	(c)	33	(c)	58	(b)	83	(a)
9	(c)	34	(c)	59	(a)	84	(b)
10	(b)	35	(c)	60	(c)	85	(c)
11	(b)	36	(b)	61	(b)	86	(c)
12	(c)	37	(c)	62	(a)	87	(d)
13	(c)	38	(a)	63	(c)	88	(c)
14	(c)	39	(a)	64	(b)	89	(b)
15	(d)	40	(c)	65	(c)	90	(c)
16	(b)	41	(c)	66	(b)	91	(b)
17	(a)	42	(c)	67	(a)	92	(c)
18	(a)	43	(b)	68	(d)	93	(c)
19	(b)	44	(a)	69	(d)	94	(c)
20	(a)	45	(a)	70	(b)	95	(c)
21	(c)	46	(a)	71	(b)	96	(a)
22	(a)	47	(c)	72	(b)	97	(c)
23	(d)	48	(d)	73	(c)	98	(d)
24	(b)	49	(c)	74	(c)	99	(d)
25	(b)	50	(b)	75	(a)	100	(c)



1. (c)

Explanation:**Kuiper Belt:**

- **The Kuiper Belt** (also known as the **Edgeworth–Kuiper belt**) is a region of the Solar System that exists beyond the eight major planets, extending from the orbit of Neptune (at 30 AU) to approximately 50 AU from the Sun. **Hence, option (c) is correct.**
- **New Horizons** is a NASA mission to study the dwarf planet Pluto, its moons, and other objects in the Kuiper Belt, a region of the solar system that extends from about 30 AU, near the orbit of Neptune, to about 50 AU from the Sun.

2. (b)

Explanation:

These are the stages in the evolution of the present atmosphere:

1. The first stage is marked by the **loss of primordial atmosphere**. The early atmosphere, with hydrogen and helium, is supposed to have been stripped off as a result of the solar winds.
2. In the second stage, the hot interior of the earth contributed to the evolution of the atmosphere. During the cooling of the earth, gases and water vapour were released from the interior solid earth. This started the evolution of the present atmosphere. The early atmosphere largely contained water vapour, nitrogen, carbon dioxide, methane, ammonia and very little of free oxygen. The process through which the **gases were outpoured from the interior is called degassing.**
3. Continuous volcanic eruptions contributed water vapour and gases to the atmosphere. As the earth cooled, the water vapour released started getting condensed. The carbon dioxide in the atmosphere got dissolved in rainwater and the temperature further decreased causing more condensation and more rains. The rainwater falling onto the surface got collected in the **depressions to give rise to oceans**. The earth's oceans were formed within 500 million years from the formation of the earth. This means, oceans are as old as 4,000 million years.

4. Finally, the composition of the atmosphere was modified by the living world through the process of photosynthesis. Sometime around 3,800 million years ago, life began to evolve. However, around 2,500-3,000 million years before the present, **the process of photosynthesis evolved.**

Hence, correct order is **2-1-4-3** i.e. loss of primordial atmosphere- degassing- Emergence of Oceans and other water bodies- Modification of composition of atmosphere through photosynthesis.

3. (a)

Explanation:**Cluster Development Programme (CDP):**

- This new programme has been announced by the **Ministry of Agriculture and Farmers Welfare (MoA&FW)**, Government of India for Horticulture Cluster Development to enhance the global competitiveness of the Indian horticulture sector. **Hence, statement 1 is correct.**
- Horticulture is the branch of plant agriculture dealing with garden crops, generally fruits, vegetables, and ornamental plants.
- Horticulture cluster is a regional/geographical concentration of targeted horticulture crops.
- **The National Horticulture Board** has been designated as the nodal agency for implementation of the CDP as a component of the Central Sector Scheme of NHB. **Hence, statement 2 is NOT correct.**
- The CDP is designed to leverage the geographical specialisation of horticulture clusters and promote integrated and market-led development of pre-production, production, post-harvest, logistics, branding, and marketing activities.
- MoA&FW has identified **53 horticulture clusters**, of which 12 have been selected for the pilot launch of the Programme.
- Based on the learnings from the pilot project, the programme will be scaled up to cover all 53 clusters.
- For the implementation of the CDP, a Government/ Public Sector entity will be appointed as a Cluster Development Agency (CDA) based on the recommendations of the State/Central Government.
- This involvement of state government bodies through CDA will ensure institutionalisation of learnings which can be replicated in future for the development of new clusters in the country.



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- The programme will benefit about 10 lakh farmers and is expected to attract an investment of Rs. 10,000 crore when implemented in all the 53 clusters.

4. (b)

Explanation:

Chinook:

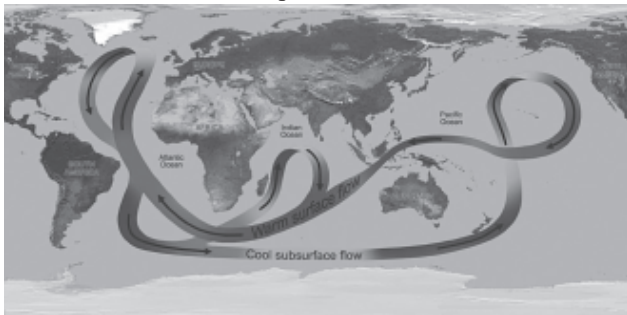
- The chinook, a native word meaning “snow eater,” belongs to a family of winds experienced in many parts of the world where long mountain chains lie more or less at right angles to the prevailing wind.
- It is a **warm and dry** westerly wind that blows down the **Rocky Mountains** into the mountains’ eastern slopes and the western prairies. **Hence, statement 1 is NOT correct.**
- The Chinook wind provides a welcome respite from the long winter chill. A strong Chinook can make large amounts of snow vanish in less than a day.
- Chinook winds have been observed to raise winter temperature, often from below -20 °C to as high as 10-20 °C for a few hours or days, then temperatures plummet to their base levels. **Hence, statement 2 is correct.**

5. (c)

Explanation:

Atlantic Meridional Overturning Circulation (AMOC)

Atlantic meridional overturning circulation (AMOC) — which is sometimes referred to as the “Atlantic conveyor belt” — is one of the Earth’s largest water circulation systems where **ocean currents move warm, salty water from the tropics to regions further north, such as western Europe and sends colder water south.**



- As warm water flows northwards in the Atlantic, it cools, while evaporation increases its salt content.

- Low temperature and high salt content increases the density of the water, causing it to sink deep into the ocean.
- The cold, dense water deep below slowly spreads southward. Eventually, it gets pulled back to the surface and warms again, and the circulation is complete. **Hence, statement 1 is correct.**
- This continual mixing of the oceans and the distribution of heat and energy around the planet contributes to the global climate.
- Atlantic Meridional Overturning Current (AMOC) **ensures the oceans are continually mixed, and heat and energy are distributed around Earth. Hence, statement 2 is correct.**

6. (d)

Explanation:

mCessation Programme - Quit Tobacco for Life:

- It is an initiative using mobile technology for tobacco cessation. India launched mCessation using text messages in 2016 as part of the government’s Digital India initiative. **Hence, statement 2 is correct.**
- The **National Tobacco Control Programme** and the **Union Ministry of Health and Family Welfare** launched the programme with support from the **World Health Organization (WHO)** and the **International Telecommunication Union’s (ITU)** ‘Be Healthy, Be mobile’ initiative’. **Hence, statement 1 is correct.**
- WHO-ITU’s ‘Be Healthy Be Mobile’ initiative, aims to reach out to tobacco users of all categories who want to quit tobacco use and support them towards successful quitting through constant text messaging on mobile phones.
- It uses two-way messaging between the individual seeking to quit tobacco use and programme specialists providing them dynamic support. **Hence, statement 3 is correct.**
- Tobacco kills almost one million persons in India every year. Tobacco is also a major risk factor for cancer, CVDs, chronic lung diseases & stroke.
- There is a huge demand for quitting tobacco use among the users as per Global Adult Tobacco Survey.



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7. (d)

Explanation:**Insolation:**

- The energy received by the earth is known as **incoming solar radiation** which is also known as insolation.
 - The amount and the intensity of insolation vary during a day, season and a year. The factors that cause these variations in insolation are:
 - (i) **The rotation of earth on its axis**
 - (ii) **The angle of inclination of the sun's rays**
 - (iii) **The length of the day**
 - (iv) The transparency of the atmosphere
 - (v) The configuration of land in terms of its aspect
- Thus, all the given factors cause variation in the amount of insolation. **Hence, statements 1, 2 and 3 are correct.**

8. (c)

Explanation:

The correct order in Nur-Sultan → Bishkek → Tashkent → Dushanbe

Hence, option (c) is correct.

9. (c)

Explanation:**Shanghai Cooperation Organisation (SCO):**

- The creation of SCO was announced on 15 June 2001 in Shanghai (China) by the Republic of Kazakhstan, the People's Republic of China, the Kyrgyz Republic, the Russian Federation, the Republic of Tajikistan, and the Republic of Uzbekistan. **Hence, statement 1 is NOT correct.**
- It is a permanent intergovernmental international organisation.
- It was preceded by the **Shanghai Five mechanism**.
- The SCO Charter was signed during the St. Petersburg SCO Heads of State meeting in June 2002 and entered

into force on 19 September 2003. **Hence, statement 2 is correct.**

- In June 2017 in Astana, the status of a full member of the Organization was granted to the Republic of India and the Islamic Republic of Pakistan.
- The Heads of State Council (HSC) is the supreme decision-making body in the SCO. It meets once a year and adopts decisions and guidelines on all important matters of the organisation.
- The SCO Heads of Government Council (HGC) meets once a year to-
 - discuss the organisation's multilateral cooperation strategy and priority areas
 - resolve current important economic and other cooperation issues
 - approve the organisation's annual budget
- The SCO's official languages are **Russian and Chinese**.
- The organisation has two permanent bodies - the **SCO Secretariat** based in Beijing and the Executive Committee of the **Regional Anti-Terrorist Structure (RATS)** based in Tashkent. **Hence, statement 3 is correct.**
- The SCO Secretary-General and the Director of the Executive Committee of the SCO RATS are appointed by the Council of Heads of State for a term of three years.
- **SCO comprises eight member states (as of November 2021), namely India, Kazakhstan, China, the Kyrgyz Republic, Pakistan, Russia, Tajikistan, and Uzbekistan. Hence, statement 4 is NOT correct.**
- SCO counts **four observer states**, namely Afghanistan, Belarus, Iran and Mongolia.
- SCO has **six dialogue partners**, namely Azerbaijan, Armenia, Cambodia, Nepal, Turkey, and Sri Lanka.

10. (b)

Explanation:**Great Barrier Reef:**

The Great Barrier Reef is the **world's largest coral reef system** composed of over 2,900 individual reefs and 900 islands stretching for over 2,300 kilometres.

It is a **World Heritage Site since 1981**.

It is located off the coast of Queensland in North-Eastern Australia. **Hence, option (b) is correct.**



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**EnVision:**

- It is a **European Space Agency (ESA)** led mission with contributions from NASA. It is likely to be launched sometime in the 2030s.
- The earliest launch opportunity for **EnVision** is **2031**, followed by 2032 and 2033.
- Once launched on an Ariane 6 rocket, the spacecraft will take about **15 months to reach Venus** and will take 16 more months to achieve orbit circularisation.
- The spacecraft will carry a range of instruments to study the planet's atmosphere and surface, monitor trace gases in the atmosphere and analyse its surface composition.
- A radar provided by NASA will help to image and map the surface.
- EnVision will follow another ESA-led mission to Venus called 'Venus Express' (2005-2014).

Hence, pair 2 is correctly matched.

13. (c)**Explanation:****Inversion of Temperature:**

- Normally, temperature decreases with increase in elevation. It is called **normal lapse rate**. Hence, **statement 1 is correct**.
- At times, the situations are reversed and the normal lapse rate is inverted. It is called **Inversion of temperature**.
- Inversion is usually of short duration but quite common. A long winter night with clear skies and still air is ideal situation for inversion.
- The heat of the day is radiated off during the night, and by early morning hours, the earth is cooler than the air above. Over polar areas, temperature inversion is normal throughout the year.
- **Surface inversion promotes stability in the lower layers of the atmosphere.** Smoke and dust particles get collected beneath the inversion layer and spread horizontally to fill the lower strata of the atmosphere. **Dense fogs in mornings** are common occurrences especially during winter season.
- This inversion commonly lasts for few hours until the sun comes up and begins to warm the earth.
- **The inversion takes place in hills and mountains due to air drainage.** Cold air at the hills and mountains, produced during night, flows under the influence of

gravity. Being heavy and dense, the cold air acts almost like water and moves down the slope to pile up deeply in pockets and valley bottoms with warm air above. This is called air drainage. **It protects plants from frost damages. Hence, statement 2 is correct.**

14. (c)**Explanation:**

- The countries that surround the Persian Gulf are: Bahrain, Iran, Iraq, Kuwait, Qatar, Saudi Arabia, and the United Arab Emirates.
- Correct 'west to east' order will be Kuwait-Bahrain-United Arab Emirates-Oman. **Hence, option (c) is correct.**

**15. (d)****Explanation:**

- Recently, the World Employment and Social Outlook: Trends 2021 report was released by the **International Labour Organisation (ILO)**. Hence, **option (d) is correct**.

Findings of the report:

- The report projects the global crisis-induced 'jobs gap' will reach 75 million in 2021, before falling to 23 million in 2022.
- The related gap in working hours, which includes the jobs gap and those on reduced hours, amounts to the equivalent of 100 million full-time jobs in 2021 and 26 million full-time jobs in 2022.



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- This shortfall in employment and working hours comes on top of persistently high pre-crisis levels of unemployment, labour underutilization and poor working conditions.
- In consequence, global unemployment is expected to stand at 205 million people in 2022, greatly surpassing the level of 187 million in 2019.
- This corresponds to an unemployment rate of 5.7 percent.
- Excluding the COVID-19 crisis period, such a rate was last seen in 2013.
- The worst affected regions in the first half of 2021 have been Latin America and the Caribbean, and Europe and Central Asia.
- Global employment recovery is projected to accelerate in the second half of 2021, provided that there is no worsening in the overall pandemic situation.
- The fall in employment and hours worked has translated into a sharp drop in labour income and a corresponding rise in poverty.
- Compared to 2019, an additional 108 million workers worldwide are now categorized as poor or extremely poor (meaning they and their families live on the equivalent of less than US\$3.20 per person per day).
- The COVID-19 crisis has also made pre-existing inequalities worse by hitting vulnerable workers harder, the report finds.
- The crisis has also hit women disproportionately. Their employment declined by 5 per cent in 2020 compared to 3.9 per cent for men. A greater proportion of women also fell out of the labour market, becoming inactive.
- Globally, youth employment fell 8.7 per cent in 2020, compared with 3.7 per cent for adults, with the most pronounced fall seen in middle-income countries.

16. (b)

Explanation:

Nature Based Solutions:

- It is Nature-based Solutions (NbS) that are defined by IUCN as “actions to protect, sustainably manage, and restore natural or modified ecosystems that address societal challenges effectively and adaptively, simultaneously providing human well-being and biodiversity benefits”. **Hence, option (b) is correct.**

The overarching goals of NbS:

- NbS are intended to support the achievement of society's development goals and safeguard human well-being in ways that reflect cultural and societal values and enhance the resilience of ecosystems, their capacity for renewal and the provision of services.

- NbS are designed to address major societal challenges, such as food security, climate change, water security, human health, disaster risk, social and economic development.

17. (a)

Explanation:

Mass Movements:

- These movements transfer the mass of rock debris down the slopes under the **direct influence of gravity**. That means, **air, water or ice do not carry debris with them** from place to place but on the other hand the debris may carry with it air, water or ice. **Hence, statement 1 is correct.**
- The movements of mass may range from **slow to rapid**, affecting shallow to deep columns of materials and include creep, flow, slide and fall.
- Gravity exerts its force on all matter, both bedrock and the products of weathering.
- So, weathering is not a prerequisite for mass movement though it aids mass movements.

18. (a)

Explanation:

- Recently, the third edition of the **Sustainable Development Goals (SDG) India Index and Dashboard 2020–21** was released by **NITI Aayog**.

SDG India Index:

- It evaluates the progress of states and Union Territories (UTs) on various parameters including health, education, gender, economic growth, institutions, climate change and environment. **Hence, statement 2 is correct.**
- It was first launched in December 2018, and now it has become the primary tool for monitoring progress on the SDGs in India. **Hence, statement 1 is correct.**
- In the Index, scores range between 0–100, the higher the score of a State/UT, the greater the distance to target achieved.
- States and UTs are classified into four categories based on their scores — **aspirant: 0–49; performer: 50–64; front-runner: 65–99; achiever: 100.**
- Currently, there are no states in the aspirant and achiever category; 15 states/UTs are in the performer category and 22 states/UTs in the front runner category.

Findings of SDG India Index 2020-21:

- The country's overall SDG score improved by 6 points from 60 in 2019 to 66 in 2020-21; on account of



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improvement in performance in providing facilities including clean water and sanitation, affordable and clean energy among others.

- While **Kerala** retained its rank as the top with a score of **75**, **Himachal Pradesh and Tamil Nadu both took the second spot** with a score of 74. Bihar, Jharkhand and Assam were the worst-performing states in this year's India index. **Hence, statement 4 is NOT correct.**
- **Chandigarh** maintained its top spot among the UTs with a score of 79, followed by Delhi (68).
- **Mizoram, Haryana and Uttarakhand** are the top gainers in 2020-21 in terms of improvement in score from 2019, with an increase of 12, 10 and 8 points, respectively.
- While in 2019, 10 states/UTs belonged to the category of front-runners (score in the range 65-99, including both), 12 more states/UTs found themselves in this category in 2020-21.
- Uttarakhand, Gujarat, Maharashtra, Mizoram, Punjab, Haryana, Tripura, Delhi, Lakshadweep, Andaman and Nicobar Islands, Jammu and Kashmir and Ladakh graduated to the category of front-runners (scores between 65 and 99, including both).

Significance:

- The index is developed in collaboration with the **United Nations in India**. **Hence, statement 3 is correct.**
- It tracks the progress of all states and UTs on **115 indicators aligned with the National Indicator Framework (NIF) of the Ministry of Statistics and Programme Implementation.**
- An index is a key tool for focused policy dialogue, formulation and implementation through development actions, in line with the global SDG framework.

19. (b)

Explanation:

River Terraces:

- River terraces are surfaces marking **old valley floor or floodplain levels**. They may be bedrock surfaces without any alluvial cover or alluvial terraces consisting of stream deposits.
- River terraces are basically products of erosion as they result due to **vertical erosion** by the stream into its own depositional floodplain. **Hence, statement 1 is NOT correct.**
- There can be a number of such terraces at different heights indicating former river bed levels.
- The river terraces may occur at the same elevation on either side of the rivers in which case they are called paired terraces.

Meanders

- In large flood and delta plains, rivers rarely flow in straight courses. Loop-like channel patterns called meanders develop over flood and delta plains.
- **Meander is not a landform but is only a type of channel pattern.** This is because of:
 - propensity of water flowing over very gentle gradients to work laterally on the banks;
 - unconsolidated nature of alluvial deposits making up the banks with many irregularities which can be used by water exerting pressure laterally;
 - coriolis force acting on the fluid water deflecting it like it deflects the wind.
 - When the gradient of the channel becomes extremely low, water flows leisurely and starts working laterally.

Hence, statement 2 is correct.

20. (a)

Explanation:



Hence, option (a) is correct.



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**21. (c)****Explanation:****World Environment Day:**

- In 1972, the UN General Assembly designated 5 June as World Environment Day (WED). **Hence, statement 1 is NOT correct.**
- WED was established on the first day of the **Stockholm Conference** on Human Environment.
- The celebration of the first World Environment Day took place under the slogan **“Only One Earth”** in 1974. **Hence, statement 2 is correct.**
- Since then, it has been celebrated every year on 5 June, engaging governments, businesses and citizens in an effort to address pressing environmental issues.
- The theme for World Environment Day 2021 was **‘Ecosystem-Restoration’**. **Hence, statement 4 is NOT correct.**
- Ecosystem restoration means preventing, halting and reversing this damage – to go from exploiting nature to healing it.
- **Pakistan** was the global host country in collaboration with the United Nations Environment Programme, for World Environment Day 2021. **Hence, statement 3 is correct.**
- WED 2021 kicked off the UN Decade on Ecosystem Restoration (2021-2030), a global mission to revive billions of hectares, from forests to farmlands, from the top of mountains to the depth of the sea.

22. (a)**Explanation:**

- **Goldilocks zone** is the area around a star where it is not too hot and not too cold for liquid water to exist on the surface of surrounding planets. **Hence, statement 1 is correct.**
- NASA’s Double Asteroid Redirection Test (DART) Mission is a planetary defense driven test of technologies for preventing an impact on earth by a hazardous asteroid. **Hence, statement 2 is NOT correct.**
- NASA’s Kepler and Transiting Exoplanet Survey Satellite (TESS) missions are tasked with locating exoplanets outside the solar system.

23. (d)**Explanation:****Soil Formation:**

- In India, soil formation is mainly related to the **parent rock material, surface relief, climate and natural vegetation**. **Hence, option (d) is correct.**

- Animals, insects and man also play an important role in soil formation.

24. (b)**Explanation:****SAGE Initiative:**

- **SAGE (Seniorcare Ageing Growth Engine)** initiative and SAGE portal to support India’s elderly has been launched by the **Ministry of Social Justice and Empowerment**. **Hence, statement 1 is NOT correct.**
- The SAGE portal will be a **“one-stop access” of elderly care products and services by credible start-ups**.
- The start-ups will be selected on the **basis of innovative products and services**, which they should be able to provide across sectors such as health, housing, care centres, apart from technological access linked to finances, food and wealth management, and legal guidance.
- The SAGE programme and SAGE portal have been launched with a view to help such persons who are interested in entrepreneurship in the field of providing services for elderly care. **Hence, statement 2 is correct.**
- The start-ups selected under SAGE will be those which will provide new innovative products and services to elderly persons in various areas like health, travel, finance, legal, housing, food among others.
- The SAGE project aims to identify, evaluate, verify, aggregate, and deliver products, solutions and services directly to the stakeholders.
- The Ministry will act as a facilitator, enabling the elderly to access the products through these identified start-ups.
- An allocation of Rs 25 crores has been made for the SAGE project in the current financial year i.e 2021-22.
- India’s elderly population is on the rise, and as per surveys, the share of elders, as a percentage of the total population in the country, is expected to increase from around 7.5% in 2001 to almost 12.5% by 2026, and surpass 19.5% by 2050.
- There is an urgent need to create a more robust eldercare ecosystem in India, especially in the post-COVID phase.

25. (b)**Explanation:**

- The Amazon rainforest is a **moist broadleaf tropical rainforest** in the Amazon biome that covers most of the Amazon basin of South America. This basin

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encompasses 7,000,000 km² (2,700,000 sq mi), of which 5,500,000 km² (2,100,000 sq mi) are covered by the rainforest.



- The majority of the forest is contained within **Brazil**, with **60% of the rainforest**, followed by **Peru** with 13%, **Colombia** with 10%, and with minor amounts in **Venezuela, Ecuador, Bolivia, Guyana, Suriname and French Guiana**.
- This region includes territory belonging to **nine nations**. Hence, **statement 1 is NOT correct while statement 2 is correct**.

26. (d)

Explanation:

Madden-Julian Oscillation:

- The Madden-Julian Oscillation (MJO) is an oceanic-atmospheric phenomenon which affects weather activities **across the globe, especially in the tropical regions**. It is characterised by disturbance of clouds and winds over **the Indian and Pacific Ocean** in which the winds moves eastward at a speed of 4-8 metres per second. Since, it involves the **eastward movement of winds**, therefore is a **traversing phenomenon** rather than stationary system. Hence, **statement 1 is NOT correct**.
- MJO was discovered in 1971 by **Dr. Roland Madden** and **Dr. Paul Julian** of the American based National Centre for Atmospheric Research (NCAR). They found that this strange phenomenon was of oscillating nature (a periodic movement between two points) and hence named it Madden-Julian Oscillation (MJO). It goes around the globe in 30-60 days on average. **It is not**

a yearly phenomenon but a monthly phenomenon. Hence, statement 2 is NOT correct.

- **The MJO consists of two parts:** one is the **enhanced rainfall (or convective)** phase and the other is the **suppressed rainfall phase**. Its journey is completed in eight phases. When it is over the Indian Ocean during the Monsoon season, it brings good rainfall over the Indian subcontinent. On the other hand, when it witnesses a longer cycle and stays over the Pacific Ocean, **MJO brings bad news for the Indian Monsoon**. Therefore, it does not always disrupts the Indian Monsoon but rather has two phases “wet phase” or “dry phase”. Hence, **statement 3 is NOT correct**.
- MJO was recently in news when the Australian Bureau of Meteorology, predicted trigger of good North-East monsoon in November due to positive phase of MJO over the Indian Ocean.

27. (c)

Explanation:

Silver Economy:

- Silver economy is the system of production, distribution and consumption of goods and services aimed at using the purchasing potential of older and ageing people and satisfying their consumption, living and health needs. Hence, **option (c) is correct**.
- The silver economy is analyzed in the field of social gerontology (study of ageing) not as an existing economic system but as an instrument of ageing policy and the political idea of forming a potential, needs-oriented economic system for an ageing population.
- Its main element is gerontechnology (Technology pertaining to aged people) as a new scientific, research and implementation paradigm.

28. (b)

Explanation:

The climate of Western Europe is a **marine climate**. Its abundant moisture comes from the **Atlantic Ocean**. The ocean moderates the temperature during the seasons, making the winter moist and mild, and the summer moist and cool. This climate primarily stays to the north of the Alps, thus considering the **Alps a climatic divide**. **Statement 1:** It is under the influence of westerlies throughout the year. Hence, **statement 1 is NOT correct**.



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Statement 2: The relatively warm waters of the North Atlantic Drift are responsible for moderating the climate of western Europe, so that winters are less cold than would otherwise be expected at its latitude. **Hence, statement 2 is correct.**

29. (b)

Explanation:

Cosmic Infrared Background Experiment-2 (CIBER-2):

- It aims to **count the number of stars** that exist in the Universe. **Hence, statement 1 is NOT correct.**
- This instrument will launch aboard a sounding rocket that will carry scientific instruments on brief trips into space before it falls back to Earth for recovery.
- While this is **not the first time** that such a mission has been undertaken, the CIBER-2 instrument has been improved upon to see **if any stars had been undercounted in the previous counting attempts.** **Hence, statement 3 is NOT correct.**
- In order to roughly estimate the number of stars in the Universe, scientists have estimated that on average **each galaxy consists of about 100 million stars**, but this figure is not exact. The figure of 100 million could easily be an underestimation, probably by a factor of 10 or more.

Counting of stars by CIBER-2:

- It will survey a patch of sky that will include dozens of clusters of galaxies. Even so, NASA notes that the **instrument will not actually count individual stars but it will instead detect the extragalactic background light**, which is all of the light that has been emitted throughout the history of the Universe. **Hence, statement 2 is correct.**
- From all of this extragalactic background light, the CIBER-2 will focus on a portion of this called **cosmic infrared background**, which is emitted by some of the most common stars.
- Essentially, this approach is aiming to look at how bright this light is to give scientists an estimate of how many of these stars are out there.
- The European Space Agency infrared space observatory Herschel also counted the number of galaxies in infrared and measured their luminosity previously. ESA says there could be 100 thousand million stars in the Milky Way alone.

30. (d)

Explanation:

Vaigai River:

- The Vaigai basin is an important basin among the **12 basins** lying between the **Cauvery and Kanyakumari**.
- This basin is bounded by the Varusanadu hills, the Andipatti hills, the Cardamom Hills and the Palani Hills on the West and by the Palk Strait and Palk Bay on the East.
- The basin is elongated in shape and **drains into Palk Bay**. The Vaigai drains an area of 7,741 sq. Km, which entirely lies in the state of Tamil Nadu.
- The Vaigai river rises on the Eastern slopes of the Varusanadu hills at an elevation of 1,200 m near Kottaimalai in the Madurai district and flows in the Northerly and North-Easterly directions up to its confluence with the Varahanadhi and then takes a turn towards the East and South-East to flow through Madurai, Sivagangai and Ramanathapuram districts.
- After traversing for about 258 Km, **river Vaigai discharges into Ramnad big tank and some other tanks**. The surplus water from the tanks finally discharges into the Palk Bay near Mandapam.
- On its way, the Vaigai receives two important tributaries namely, the **Suruliyar and the Manjalar** on its left bank, besides a large number of small streams and rivulets.
- The Suruliyar and the Manjalar, together account for nearly 20 percent of the total catchment area of the Vaigai.
- The Vaigai also receives another minor tributary namely, the Varahanadhi on its left bank below the Vaigai dam.
- The Seonath, the Jonk, the Hasdeo, the Mand, the ib, the Ong and the Tel are the main tributaries of Mahanadi which rises from the Raipur district of Chhattisgarh and flows for about 851 km before its outfall into the Bay of Bengal. **Hence, option (d) is correct.**

31. (c)

Explanation:

- The actual amount of the water vapour present in the atmosphere is known as the absolute humidity. **Hence, statement 3 is correct.**
- It is the **weight of water vapour per unit volume of air** and is expressed in terms of grams per cubic metre.



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- The ability of the air to hold water vapour depends entirely on its temperature. The absolute humidity differs from place to place on the surface of the earth.
- The percentage of moisture present in the atmosphere as compared to its full capacity at a given temperature is known as the **relative humidity**. Dry air has a higher capacity to hold water vapour than humid air. **Hence, statement 1 is NOT correct.**
- With the change of air temperature, the capacity to retain moisture increases or decreases and the relative humidity is also affected. It is greater over the oceans and least over the continents.
- The air containing moisture to its full capacity at a given temperature is said to be saturated. It means that the air at the given temperature is incapable of holding any additional amount of moisture at that stage.
- The temperature at which saturation occurs in a given sample of air is known as dew point.
- Relative humidity is an essential characteristic of weather forecasting not Absolute Humidity. **Hence statement 2 is NOT correct.**

32. (c)

Explanation:

- Temperate or cold deserts occur in temperate regions at **higher latitudes**—and therefore colder temperatures—than those at which hot deserts are found.
- These dry environments are caused by either **remoteness from the coast**, which results in low atmospheric humidity from a lack of onshore winds, or the presence of high mountains separating the desert from the coast. **Hence, statement 3 is correct.**
- They have short, moist, and moderately warm summers with fairly long, cold winters. **Hence, statement 1 is NOT correct.**
- The largest area of temperate desert lies in Central Asia, with smaller areas in western North America, south-eastern South America, and southern Australia.
- Summers are very hot and winters are very cold. The annual range of temperature is higher than in hot deserts. **Hence, statement 2 is correct.**

33. (c)

Explanation:

Organisation for the Prohibition of Chemical Weapons (OPCW):

- It is an international organization established under the Chemical Weapons Convention (CWC) in 1997 to implement and enforce the terms of the CWC. **Hence, statement 1 is NOT correct.**
- The OPCW was awarded the Nobel Peace Prize in 2013 for 'its extensive efforts to eliminate chemical weapons'. **Hence, statement 3 is NOT correct.**
- It is headquartered in Hague, Netherlands. **Hence, statement 4 is correct.**
- The OPCW, with its 193 Member States, oversees the global endeavour to permanently and verifiably eliminate chemical weapons. **Hence, statement 2 is correct.**

Functions:

- It is authorized to perform inspections to verify that signatory states are complying with the convention.
- This includes a commitment to grant inspectors full access to chemical weapons sites.
- It also performs testing of sites and victims of suspected chemical weapons attacks.
- It also provides for assistance to and protection of States attacked or threatened with chemical weapons, cooperation in the peaceful use of chemicals.
- By the 2001 Relationship Agreement between the OPCW and the UN, the OPCW reports on its inspections and other activities to the UN through the office of the Secretary-General

34. (c)

Explanation:

- The discovery of some modern hydrothermal systems in the deep oceanic realm along the mid-ocean ridges has kindled a lot of interest primarily on account of the high concentration of sulfides, base metals (Cu and Zn) and many noble metals (Au, Ag, Pd, Pt) in them. **Hence, statement 1 is correct.**
- Apart from their economic potential, sea-floor hydrothermal vents are characterized by dense biologic communities.
- Over 700 species have been reported, making the vents among the most productive ecosystems on Earth. **Hence, statement 2 is correct.**
- **Learning:** The basic paradigm of hydrothermal activity involves sub-seafloor convective circulation of seawater through permeable rocks mainly driven by the upper mantle heat sources.



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- At water depths up to 3,700 metres, hydrothermal fluids, having seeped from the ocean into subterranean chambers where they are heated by the molten rock beneath the crust, are discharged from the black smokers at temperatures up to 400°C.
- As these fluids mix with the cold surrounding seawater, metal sulphides in the water are precipitated onto the chimneys and nearby seabed.
- These sulphides, including galena (lead), sphalerite (zinc) and chalcopyrite (copper), accumulate at and just below the seafloor, where they form massive deposits that can range from several thousands to about 100 million tonnes.

35. (c)

Explanation:

- Africa is the second largest continent after Asia.
- The Equator or 0 degrees latitude runs almost through the middle of the continent. A large part of Africa lies in the Northern Hemisphere. **Hence, statement 1 is correct.**
- It is the only continent through which the Tropic of Cancer, the Equator and the Tropic of Capricorn pass. **Hence, statement 2 is correct.**



36. (b)

Explanation:

Performance Grading Index (PGI):

- It is a tool to provide insights on the status of school education in States and UTs including key levers that

drive their performance and critical areas for improvement. **Hence, statement 2 is NOT correct.**

- For States and Union Territories, it was first published in 2019 with reference year 2017-18. **Hence, statement 1 is correct.**
- PGI for 2019-20 is the third publication in this series.
- PGI 2019-20 has been released by the Department of School Education and Literacy (DoSEL), Ministry of Education. **Hence, statement 3 is NOT correct.**

Methodology:

- PGI is structured in two categories, namely, Outcomes and Governance & Management and comprises 70 indicators in aggregate with a total weightage of 1000.
- Domains under categories include: Access, Infrastructure & Facilities, Equity, Governance process.

Key Findings of PGI 2019-20:

- **State-wise Performance:**
 - It shows that 33 States and UTs have improved their PGI scores in 2019-20 compared to the previous year.
 - Andaman and Nicobar Islands, Arunachal Pradesh, Manipur, Puducherry, Punjab and Tamil Nadu have improved their overall scores by 10%.
- **Inter-state Differential:**
 - On a maximum possible score of 1000 points, the range between the States and UTs with the highest and the lowest score is more than 380 points in the year 2019-20.
- **Domain-wise Performance:**
 - Andaman and Nicobar Islands, Lakshadweep and Punjab have shown improvement of 10% or more in the 'Access' domain.
 - Thirteen states and UTs have shown improvement by 10% or more in 'Infrastructure and Facilities' while Andaman and Nicobar Islands and Odisha have improved their scores in the domain by 20% or more.
 - In 'Equity', Arunachal Pradesh, Manipur and Odisha have shown an improvement of more than 10%.
 - In the Governance Process, 19 states have shown improvement by 10% or more while Andaman and Nicobar Islands, Andhra Pradesh, Arunachal Pradesh, Manipur, Punjab, Rajasthan and West Bengal have shown improvement by at least 20%.



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37. (c)**Explanation:**

- The lava that is released during volcanic eruptions on cooling develops into igneous rocks. The cooling may take place either on reaching the surface or also while the lava is still in the crustal portion.
- Depending on the location of the cooling of the lava, igneous rocks are classified as **volcanic rocks (cooling at the surface)** and **plutonic rocks (cooling in the crust)**.

Hence, both statements 1 and 2 are NOT correct.

- The lava that cools within the crustal portions assumes different forms. These forms are called intrusive forms
 - **Batholiths:** A large body of magmatic material that cools in the deeper levels of the crust develops in the form of large domes. They appear on the surface only after the denudational processes remove the overlying materials.
 - **Lacoliths:** These are large dome-shaped intrusive bodies with a level base and connected by a pipe-like conduit from below. It resembles the surface volcanic domes of composite volcano, only these are located at greater depths. The Karnataka plateau is spotted with domal hills of granite rocks.
 - Other intrusive forms are Lapolith, Phacololith, Sills and Dykes.

38. (a)**Explanation:**

- The Coral Triangle is a marine region that spans those parts of Indonesia, Malaysia, Papua New Guinea, the Philippines, the Solomon Islands and Timor-Leste with at least 500 species of reef-building corals.

Hence, statement 1 is correct.

- The Coral Triangle encompasses portions of 2 biogeographic regions: the Indonesian-Philippines Region, and the Far South-western Pacific Region
- 76% of the world's coral species are found in the Coral Triangle, the highest coral diversity in the world.
- The epicentre of that coral diversity is found in the Bird's Head Peninsula of Indonesian Papua, which hosts 574 species (95% of the Coral Triangle, and 72% of the world's total).
- This region has more coral reef species than anywhere else on Earth but not all. Hence, statement 2 is NOT correct.

- 6 of the world's 7 marine turtle species, including the leatherback marine turtle, are found in places such as the Northern Bird's Head Peninsula / Waigeo region, Papua (Indonesia), as well as Lea region (Papua New Guinea), New Georgia (Solomon Islands) and not ALL. Hence, statement 3 is NOT correct.

39. (a)**Explanation:**

- Recently, Turkey's Sea of Marmara, which connects the Black Sea to the Aegean Sea, witnessed the largest outbreak of 'sea snout'. The sludge has also been spotted in the adjoining Black and Aegean seas. Hence, statement 1 is correct but 3 is NOT correct.
- A 'sea snout' outbreak was first recorded in the country in 2007.
- Sea snout is marine mucilage that is formed when algae are overloaded with nutrients as a result of water pollution combined with the effects of climate change. Hence, statement 2 is correct.
- The nutrient overload occurs when algae feast on warm weather caused by global warming.
- It looks like a viscous, brown and foamy substance.

Threats:

- This outbreak has caused mass deaths among the fish population, and also killed other aquatic organisms such as corals and sponges.
- It is now covering the surface of the sea and has also spread to 80-100 feet below the surface which eventually can collapse to the bottom and cover the sea floor.
- This outbreak is affecting the livelihoods of fishermen as the sludge is getting collected in their nets, making them so heavy that they break or get lost. Moreover,


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the mucilage coating the strings, makes the nets visible to fish and keeps them away.

- Also, the sea snots can cause an outbreak of water-borne diseases such as cholera in cities like Istanbul.

Steps taken:

- The entire Sea of Marmara will be turned into a protected area.
- Moreover, steps are being taken to reduce pollution and improve the treatment of waste water from coastal cities and ships.
- Turkey's biggest maritime clean-up operation is being launched and called on local residents, artists and NGOs to join hands to extend assistance.

40. (c)

Explanation:

Karst is a terrain with a characteristic relief and drainage arising mainly due to higher solubility of rock in natural water than is found elsewhere. **Hence, statement 2 is correct.**

It is a dry, upland landscape with underground drainage instead of surface streams. It is so named after a province of Yugoslavia on the Adriatic Sea coast where such formations are most noticeable.

Conditions Essential for Full Development of Karst Topography:

1. Presence of soluble rocks, preferably limestone at the surface or sub-surface level.
 2. The rainfall should be neither too high nor too low.
 3. There should be a perennial source of water.
- Hence, statement 1 is NOT correct.**

Karst Landforms:

- Karren/Lapies
- **Cavern:** This is an **underground cave** formed by water action by various methods in a limestone or chalk area. In India, such caves can be seen in Bastar, Dehradun, Shillong plateau.
- **Arch/Natural Bridge:** When a part of the cavern collapses the portion which keeps standing forms an arch.
- **Sink Hole/Swallow Hole:** Sink holes are funnel-shaped depressions, these holes are developed by enlargement of the cracks found in such rocks, as a result of continuous solvent action of the rainwater.

- **Karst Window:** When a number of adjoining sink holes collapse, they form an open, broad area called a karst window.
- **Sinking Creeks/Bogas:** In a valley, the water often gets lost through cracks and fissures in the bed. These are called sinking creeks, and if their tops are open, they are called bogas.
- **Dolines:** These are small depressions dotting a karst landscape.
- **Uvala:** A number of adjoining dolines may come together to form a large depression called uvala. **Hence, statement 3 is correct.**
- **Polje/Blind Valley:** A number of uvalas may coalesce to create a valley called polje which is actually a flat-floored depression. If the streams lose themselves in these valleys, then these are called blind valleys.
- **Dry Valley/Hanging Valley/Bourne:** Sometimes, a stream cuts through an impermeable layer to reach a limestone bed. It erodes so much that it goes very deep. The water table is also lowered. Now the tributaries start serving the subterranean drainage and get dried up. These are dry valleys or bournes. Lack of adequate quantities of water and reduced erosion leaves them hanging at a height from the main valley. Thus, they are also referred to as hanging valleys.
- **Hums:** These are curved relicts of limestone rocks after erosion.
- Stalactite and Stalagmite

41. (c)

Explanation: Seismic Waves

- P-waves can travel through solids and liquids whereas S-waves can only travel through solids. **Hence, statement 1 is NOT correct.**
- P-waves are longitudinal waves i.e. they vibrate along the same direction as it travels. Whereas, **S-waves are transverse waves. Hence, statement 2 is NOT correct.**
- P-waves are the first waves detected by a seismograph. **Hence, statement 3 is correct.**

42. (c)

Explanation:

- Interpol has launched a new global database named "I-Familia" to identify missing persons through family DNA and help the police solve cold cases in member countries. **Hence, option (c) is correct.**



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**I-Familia:**

- The first of its kind, I-Familia is a global database for identifying missing persons based on international DNA (deoxyribonucleic acid) kinship matching.
- The database seeks to identify missing persons or unidentified human remains when direct comparison is not possible, by using DNA samples from family members instead.
- Family members must give their consent for their data to be used for international searching.
- It builds on Interpol's long-standing success in direct DNA matching.

Working:

- **I-Familia has three components:**
 - I. A dedicated global database to host the DNA profiles provided by relatives, held separately from any criminal data.
 - II. DNA matching software called Bonaparte.
 - III. Interpretation guidelines, produced by Interpol, to efficiently identify and report potential matches.
- In the event of a match, notifications are sent to the countries that supplied the DNA profile from the unidentified body and from the family respectively.

Interpol:

- The International Criminal Police Organization (Interpol) is an intergovernmental organization that helps coordinate the police force of 194 member countries.
- Each of the member countries hosts an Interpol National Central Bureau (NCB). This connects their national law enforcement with other countries and with the General Secretariat.
- The Central Bureau of Investigation (CBI) is designated as the National Central Bureau of India.
- Interpol is headquartered in Lyon, France.
- Interpol Notices are international requests for cooperation or alerts allowing police in member countries to share critical crime-related information.

43. (b)**Explanation:**

- In June 2021, Prime Minister announced the 'Young, Upcoming and Versatile Authors' (YUVA) scheme, a mentorship programme to train young authors.

- The scheme is aimed at training 75 aspiring writers below 30 years, who are ready to express themselves and project India and its culture and literature globally. **Hence, statement 1 is NOT correct.**

- **Through this scheme -**

- The government aims to bring reading and authorship as a preferred profession at par with other job options.
- It is also expected to impart a positive psychological push to the young minds amidst the impact of the Covid-19 pandemic on the mental health of children.
- The National Book Trust, India (under BP Division, Ministry of Education, GOI) will be the implementing agency and it will ensure phase-wise execution of the Scheme under well-defined stages of mentorship.

PHASE I – TRAINING (3 MONTHS)

- The NBT will organise a Two-Week Writers' Online Programme for the selected candidates.
- During which the young authors will be trained by two eminent authors/mentors from NBT's panel of accomplished authors and writers.
- After the completion of the two-week writers' Online Programme, the authors will be trained for 2-Weeks at various online/on-site National Camps organised by NBT.

PHASE II – PROMOTION (3 MONTHS)

- The young authors will get to expand their understanding and hone their skills through interaction at various international events.
- At the end of mentorship a consolidated scholarship of ₹50,000 per month for a period of 6 months ($50,000 \times 6 = ₹3 \text{ Lakh}$) per author will be paid under the Mentorship Scheme. **Hence, statement 2 is correct.**
- A book or a series of books written by the young authors will be published by NBT as the outcome of the mentorship programme.
- A royalty of 10% will be payable to the authors on successful publications of their books at the end of the Mentorship Program.
- Their published books will be translated into other Indian languages ensuring the exchange of culture & literature between different states & thereby promoting Ek Bharat Shreshtha Bharat.

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**44. (a)****Explanation:**

Mid-ocean ridges along **divergent plate boundaries**, where new ocean floor is created as the Earth's tectonic plates spread apart. As the plates separate, some molten rock rises to the seafloor, producing enormous volcanic eruptions of basalt, and building the longest chain of volcanoes in the world. Because most of these eruptions occur deep under the water, they often go unnoticed. This uplifting of the ocean floor occurs when convection currents rise in the mantle beneath the oceanic crust and create magma where two tectonic plates meet at a divergent boundary. **Hence, option (a) is correct.**

45. (a)**Explanation:**

- Recently, the Assam government notified Dehing Patkai as the 7th National Park of the state.
- It was created shortly after Raimona reserve forest in western Assam's Kokrajhar district was upgraded to a National Park (6th) on the occasion of World Environment Day.

Dehing Patkai National Park:

- It is located within the larger Dehing Patkai Elephant Reserve, which spreads across the coal- and oil-rich districts of Upper Assam (Dibrugarh and Tinsukia districts).
- The oldest refinery of Asia in Digboi and 'open cast' coal mining at Lido are located near the sanctuary.
- The Dehing Patkai Wildlife Sanctuary is also known as the Jeypore Rainforest.
- Dehing is the name of the river that flows through this forest and Patkai is the hill at the foot of which the sanctuary lies.
- Rare fauna found in the region include Chinese pangolin, flying fox, wild pig, sambar, barking deer, gaur, serow and Malayan giant squirrels.
- It is the only sanctuary in India which is home to seven different species of wild cats - tiger, leopard, clouded leopard, leopard cat, golden cat, jungle cat and marbled cat.
- Assamese macaque, a primate found in the forest, is in the red list of Near Threatened species.
- It has the highest concentration of the rare endangered White Winged Wood Duck.

National Parks in Assam:

- Assam now has the third most National Parks after the 12 in Madhya Pradesh and nine in the Andaman and Nicobar Islands.
- The seven National Parks in the State are Dehing Patkai, Raimona, Kaziranga, Manas, Nameri, Orang, and Dibrus-Saikhowa. **Hence, option (a) is correct.**
- Kaziranga and Manas are UNESCO World Heritage Sites. They are also tiger reserves along with Nameri and Orang.

46. (a)**Explanation:**

- White dwarfs are stars that have burned up all of the hydrogen they once used as nuclear fuel. Such stars have very high density. When the hydrogen, used as fuel, vanishes and fusion slows, gravity causes the star to collapse in on itself into **white dwarfs**. **Hence, statement 1 is correct.**
- It must be noted that **not all white dwarfs** cool and transform into black dwarfs. Those white dwarfs which have **enough mass** reach a level called the **Chandrasekhar Limit**. At this point, the pressure at its centre becomes so great that the star will detonate in a **thermonuclear supernova**. **Hence, statement 2 is NOT correct.**

47. (c)**Explanation:**

- The Baltic Sea is surrounded by nine countries: Denmark, Germany, Poland, Lithuania, Latvia, Estonia, Russia, Finland and Sweden. **Hence, option (c) is correct.**

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**48. (d)****Explanation:****International Yoga Day**

- June 21 is globally celebrated as International Yoga Day, recognising the many benefits of the ancient Indian practice of yoga.
- The year 2021 marked the 7th annual International Yoga Day. **Hence, statement 1 is correct.**
- In India, this day was celebrated by the Ministry of Culture at 75 cultural heritage locations across the country.
- The extensive drive (Yoga at 75 cultural heritage) was titled “Yoga, An Indian Heritage”, and it was part of India’s “Azadi ka Amrit Mahotsav” campaign.
- Some places in the list of 75 sites were the Agra Fort in Uttar Pradesh, Shanti Stupa in Ladakh, Ellora Caves in Maharashtra and Nalanda in Bihar, Rajiv Lochan Temple, Raipur, Sabarmati Ashram in Gujarat, and Akhnoor Fort in Jammu.
- The United Nations theme for this year’s Yoga Day was “Yoga for well-being”, which takes into account how the practice can promote the holistic health of every individual. **Hence, statement 2 is correct.**
- The UN proclaimed June 21 as International Day of Yoga by passing a resolution on December 11, 2014 during the 69th session of the General Assembly. **Hence, statement 3 is correct.**
- The first Yoga Day celebrations were held in 2015 at Rajpath in New Delhi.

Yoga:

- It is an ancient physical, mental and spiritual practice that originated in India.
- The word ‘yoga’ is derived from Sanskrit and means to join or to unite, symbolizing the union of body and consciousness.
- WHO has asked its member states to practice Yoga and has included it in its Global Action Plan for physical activity 2018-30.

49. (c)**Explanation:**

- Water vapour in the atmosphere can condense or deposit to form clouds and precipitation which falls to earth as rain, snow, or hail.

- There is nearly four times as much precipitation over oceans than precipitation over land. **Hence, statement 1 is NOT correct.**
- Out of the total water present on the earth and in its atmosphere, it is found to be more in freshwater lakes than in the soil water. **Hence, statement 2 is NOT correct.**
- An important principle concerning humidity states that the maximum quantity of water vapour an air parcel can contain is dependent on the air temperature itself. **Hence, statement 3 is correct.**
- Warm air can contain more water vapour than cold air. Air at room temperature (20°C, 68°F) can contain about three times as much water vapour as freezing air (0°C, 32°F).

50. (b)**Explanation:**641, 1st floor, Mukherjee
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Hence, option (b) is correct.

51. (b)

Explanation:

- Recently, a major temple in Assam signed a memorandum of understanding with two green NGOs for long-term conservation of the rare freshwater black softshell turtle (*Nilssononia nigricans*).
- A vision document 2030 was also launched to raise at least 1,000 black softshell turtles by 2030.

Black Softshell Turtle:

- They look almost the same as the Indian peacock softshell turtle (*Nilssononia hurum*), which is classified as Endangered in the IUCN Red List.
- It is a freshwater species and there are 29 species of freshwater turtles and tortoises found in India.
- Until its sightings along the Brahmaputra River's drainage in Assam, it was thought to be 'extinct in the wild' and confined only to ponds of temples in northeastern India and Bangladesh. **Hence, statement 2 is correct.**
- In 2021, the International Union for Conservation of Nature had listed the turtle as 'critically endangered'. **Hence, statement 1 is NOT correct.**

- It does not enjoy legal protection under the Indian Wildlife (Protection) Act of 1972 although it has traditionally been hunted for its meat and cartilage, traded in regional and international markets. **Hence, statement 3 is NOT correct.**

52. (c)

Explanation:



- Out of the given options the Tropic of Capricorn does not pass through Indonesia. **Hence, option (c) is correct.**

53. (b)

Explanation:

Ocean currents can also be classified based on temperature: cold currents and warm currents:

- Cold currents bring cold water into warm water areas. These currents are usually found on the west coast of the continents in the low and middle latitudes (true in both hemispheres) and on the east coast in the higher latitudes in the Northern Hemisphere. **Hence, statement 1 is NOT correct.**
- Warm currents bring warm water into cold water areas and are usually observed on the east coast of continents in the low and middle latitudes (true in both hemispheres). In the northern hemisphere they are found on the west coasts of continents in high latitudes. **Hence, statement 2 is NOT correct.**

Effects of Ocean Currents

- West coasts of the continents in tropical and subtropical latitudes (except close to the equator) are bordered by cool waters. Their average temperatures are relatively low with a narrow diurnal and annual ranges. There is fog, but generally the areas are arid.



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- West coasts of the continents in the middle and higher latitudes are bordered by warm waters which cause a distinct marine climate. They are characterized by cool summers and relatively mild winters with a narrow annual range of temperatures.
- Warm currents flow parallel to the east coasts of the continents in tropical and subtropical latitudes. This results in warm and rainy climates. These areas lie in the western margins of the subtropical anti-cyclones.
- The mixing of warm and cold currents help to replenish the oxygen and favour the growth of planktons, the primary food for fish population. The best fishing grounds of the world exist mainly in these mixing zones. **Hence, statement 3 is correct.**

54. (b)

Explanation:

- The term gain-of-function research describes a type of research that modifies a biological agent so that it confers new or enhanced activity to that agent. Some scientists use the term broadly to refer to any such modification. **Hence, option (b) is correct.**
- In virology, gain-of-function research involves deliberately altering an organism in the lab, altering a gene, or introducing a mutation in a pathogen to study its transmissibility, virulence and immunogenicity.
- This is done by genetically engineering the virus and by allowing them to grow in different growth mediums, a technique called serial passage.
- Serial Passage refers to the process of growing bacteria or a virus in iterations. For instance, a virus may be grown in one environment, and then a portion of that virus population can be removed, and put into a new environment.
- The Wuhan Institute of Virology was said to have conducted gain-of-function research on coronaviruses which may possibly have caused the lab-leak origin of the SARS-CoV-2 (Covid-19 pandemic).

Significance:

- This would allow researchers to study potential therapies and ways to control the disease better in future.
- Gain-of-function studies, which enhance viral yield and immunogenicity (relating to immune response), are required for vaccine development.

Issues involved:

- It involves manipulations that make certain pathogenic microbes more deadly or more transmissible.

- There is also 'loss-of-function research, which involves inactivating mutations, resulting in a significant loss of original function, or no function to the pathogen.
- It reportedly carries inherent biosafety and biosecurity risks and is thus referred to as 'dual-use research of concern' (DURC).

55. (c)

Explanation:

- The rotation of the earth about its axis affects the direction of the wind. This force is called the Coriolis force after the French physicist who described it in 1844. **Hence, statement 1 is correct.**
- It deflects the wind to the right direction in the northern hemisphere and to the left in the southern hemisphere. The deflection is more when the wind velocity is high. **The Coriolis force is directly proportional to the angle of latitude. It is maximum at the poles and is absent at the equator. Hence, statement 2 is correct.**
- The Coriolis force acts perpendicular to the pressure gradient force. The pressure gradient force is perpendicular to an isobar. The higher the pressure gradient force, the more is the velocity of the wind and the larger is the deflection in the direction of wind. As a result of these two forces operating perpendicular to each other, in the low-pressure areas the wind blows around it.
- At the equator, the Coriolis force is zero and the wind blows perpendicular to the isobars. The low pressure gets filled instead of getting intensified. That is the reason why tropical cyclones are not formed near the equator.

56. (c)

Explanation:



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Hence, option (c) is correct.

57. (c)

Explanation:

- Recently, the Department of Biotechnology (DBT) issued a Special Call for the North East Region as a part of its Mission Programme “Biotech-KISAN”.

Biotech-Krishi Innovation Science Application Network (Biotech-KISAN) Scheme:

- Launched in 2017, it is a Department of Biotechnology, Ministry of Science and Technology initiative that empowers farmers, especially women farmers. **Hence, statement 1 is NOT correct.**
- It aims to understand the problems of water, soil, seed and market faced by the farmers and provide simple solutions to them. **Hence, statement 3 is correct.**
- It is for farmers, developed by and with farmers, it empowers women, impacts locally, connects globally, is pan-India, has a hub-and spoke model and stimulates entrepreneurship and innovation in farmers. **Hence, statement 2 is NOT correct.**
- It was launched for agriculture innovation with an objective to connect science laboratories with the farmers to find out innovative solutions and technologies to be applied at farm level.

Progress:

- So far 146 Biotech-KISAN Hubs have been established covering all 15 agroclimatic zones and 110 Aspirational Districts in the country.
- It has benefitted over two lakhs farmers by increasing their agriculture output and income.
- Over 200 entrepreneurship have also been developed in rural areas.

Special call for North East Region:

- It focuses on the NE Region as it is predominantly agrarian with 70% of its workforce engaged in agriculture and allied sector for livelihood.
- The region produces merely 1.5 percent of the country’s food grain and continues to be a net importer of food grains even for its domestic consumption.
- The NER has untapped potential to enhance the income of the farming population.
- The scheme will be implemented in NER with the objective of linking available innovative agriculture technologies to the farm with the small and marginal farmers, specially women farmers of the region.



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- The Hubs in NER will collaborate with the top scientific institutions across the country as well as in the states for demonstrations of technologies and training of farmers.

58. (b)

Explanation:

The rate of evaporation increases with—

1. **An increase in surface area:** Evaporation is a surface phenomenon. If the surface area is increased, the rate of evaporation increases.
2. **An increase in temperature:** With the increase of temperature, more number of particles get enough kinetic energy to go into the vapour state.
3. **A decrease in humidity:** If the amount of water in the air is already high, the rate of evaporation decreases.
4. **An increase in wind speed:** With the increase in wind speed, the particles of water vapour move away with the wind, decreasing the amount of water vapour in the surrounding.

Hence, option (b) is correct.

59. (a)

Explanation:

- **Evergreen forests:** These forests are found in the western slope of the Western Ghats, hills of the northeastern region and the Andaman and Nicobar Islands.
 - They are found in warm and humid areas with an annual precipitation of over 200 cm and mean annual temperature above 22°C. **Hence, option (a) is correct.**
 - Tropical evergreen forests are well stratified, with layers closer to the ground and are covered with shrubs and creepers, with short structured trees followed by tall variety of trees. In these forests, trees reach great heights up to 60 m or above.
 - There is no definite time for trees to shed their leaves, flowering and fruition.
 - As such these forests appear green all year round.
 - Species found in these forests include rosewood, mahogany, aini, ebony, etc.

Semi evergreen forests are found in the less rainy parts of these regions.

- Such forests have a mixture of evergreen and moist deciduous trees. The under growing climbers provide an evergreen character to these forests.

- Main species are white cedar, hollock and kail.

Tropical moist deciduous forests are more pronounced in the regions which record rainfall between 100-200 cm.

- These forests are found in the northeastern states along the foothills of Himalayas, eastern slopes of the Western Ghats and Odisha.
- Teak, sal, shisham, hurra, mahua, amla, semul, kusum, and sandalwood etc. are the main species of these forests.

Littoral and Swamp Forests: India has a rich variety of wetland habitats. About 70 percent of this comprises areas under paddy cultivation.

- The total area of wetland is 3.9 million hectares.
- They consist of a number of salt-tolerant species of plants. Crisscrossed by creeks of stagnant water and tidal flows, these forests give shelter to a wide variety of birds.

60. (c)

Explanation:

- It has been found that the illegal cultivation of Herbicide Tolerant (HT) Bt cotton has seen a huge jump as the sale of illegal seed packets has more than doubled from 30 lakh in 2020 to 75 lakh in 2021.

Bt Cotton:

- It is the only transgenic crop that has been approved by the Centre for commercial cultivation in India. **Hence, statement 3 is NOT correct.**
- It has been genetically modified (GM) to produce an insecticide to combat the cotton bollworm, a common pest.

HTBt Cotton:

- This variant adds another layer of modification, making the plant resistant to the herbicide glyphosate, but has not been approved by regulators. **Hence, statement 2 is correct.**
- Fears include glyphosate having a carcinogenic effect, as well as the unchecked spread of herbicide resistance to nearby plants through pollination, creating a variety of superweeds.



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- It saves cost as there is a shortage of the labour needed to do at least two rounds of weeding for Bt cotton.
- Scientists are also in favour of this crop, and even the WHO has said it does not cause cancer.

Issues with HTBT Cotton:

- As it is not approved by the Genetic Engineering Appraisal Committee (GEAC), illegal sale takes place in Indian markets. **Hence, statement 1 is NOT correct.**
- There is no accountability of the quality of seed, it pollutes the environment, industry is losing legitimate seed sale and the government also loses revenue in terms of tax collection.
- It will not only decimate small cotton seed companies but also threatens the entire legal cotton seed market.

Genetic Engineering Appraisal Committee:

- It functions in the Ministry of Environment, Forest and Climate Change (MoEF&CC).
- **It is responsible for -**
 - appraisal of activities involving large scale use of hazardous microorganisms and recombinants in research and industrial production from the environmental angle.
 - appraisal of proposals relating to release of genetically engineered organisms and products into the environment including experimental field trials.
- It is chaired by the Special Secretary/Additional Secretary of MoEF&CC and co-chaired by a representative from the Department of Biotechnology (DBT).
- Presently, it has 24 members and meets every month to review the applications.

61. (b)

Explanation:



- From the above image, it is clear that among the given options, Serbia and Moldova do not share their border with the Black Sea. **Hence, option (b) is correct.**

62. (a)

Explanation:

- The term 'basket of eggs' topography refers to drumlin which is a depositional landform formed by glaciers. **Hence, option (a) is correct.**
- They are forms of **rounded hummocks** resulting from the deposition till which look like an inverted boat or spoon.
- When they occur in a cluster they look like a **basket of eggs**.

63. (c)

Explanation:

- The United Nations Global Compact is a non-binding United Nations pact to encourage businesses worldwide to adopt sustainable and socially responsible policies and to report on their implementation. **Hence statement 2 is NOT correct.**
- It is a principle-based framework for businesses, stating ten principles in the areas of human rights, labour, the environment, and anti-corruption.
- The CEO Water Mandate is a special initiative of the UN Secretary-General and the UN Global Compact, implemented in partnership with the Pacific Institute. It was established in 2007. It aims to demonstrate the commitment and efforts of companies to better their water and sanitation agendas as part of long-term Sustainable Development Goals. **Hence statements 1 and 3 are correct.**
- Under the Global Compact, companies are brought together with UN agencies, labor groups and civil society. Cities can join the Global Compact through the Cities Programme.
- India is also a part of the UN Global Compact.

64. (b)

Explanation:

Mediterranean Climate: It is characterised by three distinct features:

- Wet winter and dry summer season
 - Warm and hot summers, and mild winters
 - Abundant sunshine throughout the year
- Hence statement 1 is NOT correct.**



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- The Mediterranean climate or biome is also called as sclerophyll biome because of the development of special features and characteristics in the dominant trees and shrubs to adapt to the typical environment condition of this climate -dry summer and wet winter.
- **Location:** This climate has developed between 30° - 40° (sometimes upto 45°) latitude in both the hemisphere.
- **This climatic region includes-**
 - A. European, Asiatic and African land bordering Mediterranean sea
 - B. Central and southern California in the US.
 - C. Central Chile in South America
 - D. Cape town area of South Africa
 - E. Coastal zones of southern and south-western Australia.
- **Causes:** The Mediterranean climate owes its origin to the seasonal shifting of pressure belt due to northward and southward migration of the sun.
- Thus, these areas come under the domain of westerlies in winter season . Since the westerlies are moisture laden because they come from over the oceans and are associated with temperate cyclones, they give sufficient rains during winter season.
- On the other hand, they come under the influence of sub-tropical high pressure belt and associated anti-cyclones during summer season and hence there is no rainfall. **Hence statement 2 is correct.**

65. (c)

Explanation:

The velocity and direction of the wind are the net result of the wind generating forces. The winds in the upper atmosphere, 2-3 km above the surface, are free from frictional effect of the surface and are controlled by the pressure gradient and the Coriolis force. When isobars are straight and when there is no friction, the pressure gradient force is balanced by the Coriolis force and the resultant wind blows parallel to the isobar. This wind is known as the geostrophic wind. **Hence, both statements 1 and 2 are correct.**

66. (b)

Explanation:

- Recently, the United Airlines of USA has announced plans to buy 15 new supersonic airliners and “return supersonic speeds to aviation” in the year 2029. The

Overture aircraft from Boom Supersonic, with the planes able to fly at Mach 1.7 faster than the speed of sound, will be purchased. The new supersonic “Overture” aircraft will become the world’s fastest commercial airliner, reducing travel time by about half of today’s planes. **Hence statement 1 is correct.**

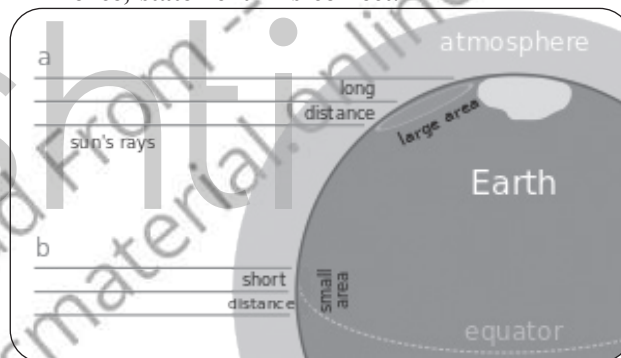
- The company has claimed to produce an eco-friendly aircraft with “net-zero carbon emissions”, set to fly with 100% sustainable aviation fuel (SAF). It aims for “zero overland noise.” Thus it will ensure that no sonic boom reaches the surface(land) where people live. **Hence statement 2 is correct.**
- The costs of making “sustainable” supersonic planes are extremely high. It will be equipped with advanced aerodynamics and carbon composite materials so that the aim of sustainability can be achieved. **Hence statement 3 is NOT correct.**

67. (a)

Explanation:

- The energy received by the earth is known as incoming solar radiation which in short is termed as insolation.
- Due to the Earth’s inclination, the mid-day sun is almost over the head within the tropics but sun’s rays reaches the earth at an angle outside the tropics. Temperature thus diminishes from equatorial regions to the poles.

Hence, statement 1 is correct.



(The effect of latitude on solar insolation. This shows why temperature are lower in higher latitudes than in the tropics.)

- Maximum insolation is received over the subtropical deserts, where the cloudiness is the least. Equator receives comparatively **less** insolation than the tropics. In winter, the middle and higher latitudes receive less radiation than in summer. **Hence, statement 2 is NOT correct.**



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**68. (d)****Explanation:**

Savanna climate is a zone of transitional between the equatorial forest and the trade wind hot deserts. It is confined within the tropics. The belt includes West African Sudan, and then curves southward into east Africa and southern Africa north of the tropic of Capricorn. **Hence, statement 1 is NOT correct.**

It is best developed in Sudan where the dry and wet seasons are most distinct, hence its name the Sudan climate. **Hence, statement 2 is NOT correct.**

In South America there are two distinct regions of Savanna:

- The Llanos a vast tropical grassland plain situated to the east of the Andes in Colombia and Venezuela, in northwestern South America. It is an ecoregion of the flooded grasslands and savannas biome.
- The Campos, grassland with few trees or shrubs except near streams, lies between 24°S and 35°S; it includes parts of Brazil, Paraguay and Argentina, and all of Uruguay.

In Australia the Savanna is located south of the monsoon strip running from west to east north of the tropic of Capricorn.

69. (d)**Explanation:**

- The phenomenon of Gravitational Lensing occurs when a huge amount of matter, such as a massive galaxy, cluster of galaxies or a black hole, creates a gravitational field that distorts and magnifies the light from objects behind it. It is based on Einstein's theory of general relativity (Mass bends light). **Hence, statements 1 and 2 are correct.**
- USA's space agency NASA is planning to launch the James Webb Space Telescope (JWST) on 22nd December 2021, which will use the natural phenomenon of "gravitational lensing" to carry out astronomical observations. **Hence, statement 3 is correct.**

70. (b)**Explanation:**

- Recently, the International Criminal Police Organization (Interpol) through its Operation Pangea XIV targeted the sale of fake medicines and products online. **Hence, statement 1 is NOT correct.**

- Operation Pangea is a well-established international effort of Interpol to disrupt the online sale of counterfeit and illicit health products. Just as importantly, Pangea works to raise awareness of the risks associated with buying medicines from unregulated websites. The first Operation Pangea was conducted in 2008. **Hence, statement 2 is correct.**

71. (b)**Explanation:**

- The Generalised Scheme of Preference Plus (GSP+) is a type of European Union's Generalised Scheme of Preference rules allowing exporters from all the developing countries to pay less or no duties on their exports to the European Union. **Hence, statement 1 is NOT correct.**
- GSP+ is also a special incentive arrangement for sustainable development and good governance. It slashes these same tariffs (that under standard GSP) to 0% for vulnerable low and lower-middle-income countries that implement international conventions related to human rights, labour rights, protection of the environment, and good governance. Beneficiaries of GSP+ are Armenia, Bolivia, Cabo Verde, Kyrgyzstan, Mongolia, Pakistan, Philippines, and Sri Lanka. **Hence, statement 2 is correct.**
- Recently, a resolution was adopted by the European Parliament, urging the European Union (EU) Commission to consider the temporary withdrawal of the Generalised Scheme of Preferences Plus (GSP+) status given to Sri Lanka.

72. (b)**Explanation:**

- According to the World Competitiveness Yearbook (WCY), India maintained 43rd rank on the annual World Competitiveness Index, 2021.
- World Competitiveness Index measures the prosperity and competitiveness of countries by examining four factors-
 - I. Economic performance
 - II. Government efficiency
 - III. Business efficiency
 - IV. Infrastructure



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Among the four indices used, India's ranking in government efficiency increased to 46 from 50 a year ago while its ranking in other parameters such as economic performance (37th), business efficiency (32nd), and infrastructure (49th) remained the same. **Hence, option (b) is correct.**

73. (c)

Explanation:

- Hurricane is a tropical cyclone that occurs in the **Atlantic Ocean** and the **northeastern Pacific Ocean**. **Hence, statement 1 is NOT correct.**
- These are formed over the **warm ocean waters near the equator**. **Hence, statement 2 is correct.**
- These typically form between **5 to 15 degrees latitude north and south of the equator**. **Hence, statement 3 is correct.**
- Hurricanes are one of nature's **most destructive storms** as they bring heavy rains, deadly waves and fierce winds. **Hence, statement 4 is NOT correct.**

74. (c)

Explanation:

- The scientists at Event Horizon Telescope Project have released the first-ever image of a **Black Hole** recently. The term 'black hole' was coined in the mid-1960s by American Physicist **John Archibald Wheeler**. **Hence, statement 1 is NOT correct.**
- It refers to a point in space where matter is so compressed as to create a gravity field from which even light cannot escape. **Hence, statement 2 is correct.**
- Black-holes were theorized by Albert Einstein in 1915. **Hence, statement 3 is NOT correct.**

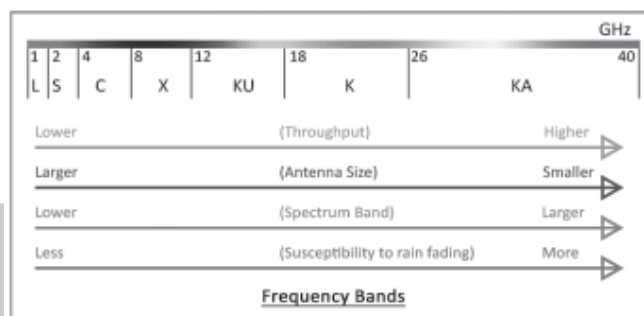
75. (a)

Explanation:

- Recently, the India Meteorological Department (IMD) announced that it will install seven new doppler radars in Maharashtra, including Mumbai in 2021. Doppler radars of varying frequencies — S-band, C-band, and X-band — are commonly used by the IMD to detect and track the movement of weather systems, cloud bands and gauge rainfall over its coverage area of about 500 kms.
- Based on the Doppler principle the radar is designed to improve precision in long-range weather forecasting

and surveillance using a parabolic dish antenna and a foam sandwich spherical radome. The smaller wavelength (higher frequency) radar is more sensitive and can detect smaller particles. **Hence, statement 1 is correct while statement 2 is NOT correct.**

- **Types Of Doppler Radars:** Doppler radar can be divided into several different categories according to the wavelength which are L,S,C,X,K.



○ **X-band radar:**

- ◆ It operates on a wavelength of 2.5-4 cm and a frequency of 8-12 GHz. Because of the smaller wavelength, the X band radar is more sensitive and can detect smaller particles.
- ◆ It is used to detect thunderstorms and lightning.

○ **C-band radars:**

- ◆ It operates on a wavelength of 4-8 cm and a frequency of 4-8 GHz. Because of the wavelength and frequency, the dish size does not need to be very large.
- ◆ The signal is more easily attenuated, so this type of radar is best used for short range weather observation.

- It guides at the time of cyclone tracking.

○ **S-band radars:**

- ◆ It operates on a wavelength of 8-15 cm and a frequency of 2-4 GHz. Because of the wavelength and frequency, S band radars are not easily attenuated.
- ◆ This makes them useful for near and far range weather observation.

76. (a)

Explanation:

- The El Niño-Southern Oscillation (ENSO) is a recurring climate pattern involving changes in the temperature of waters in the central and eastern tropical Pacific Ocean.



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- On periods ranging from about three to seven years, the surface waters across a large swath of the tropical Pacific Ocean warm or cool by anywhere from 1°C to 3°C, compared to normal. **Hence, statement 1 is correct.**
- This oscillating warming and cooling pattern, referred to as the ENSO cycle, directly affects rainfall distribution in the tropics and can have a strong influence on weather in parts of the world.
- El Niño and La Niña are the extreme phases of the ENSO cycle; between these two phases is a third phase called ENSO-neutral.
 - El Niño:** A warming of the ocean surface, or above-average sea surface temperatures (SST), in the central and eastern tropical Pacific Ocean. Over Indonesia, rainfall tends to become reduced while rainfall increases over the central and eastern tropical Pacific Ocean. The low-level surface winds, which normally blow from east to west along the equator ("easterly winds"), instead weaken or, in some cases, start blowing the other direction (from west to east or "westerly winds"). In general, the warmer the ocean temperature anomalies, the stronger the El Niño (and vice-versa).
 - La Niña:** A cooling of the ocean surface, or below-average sea surface temperatures (SST), in the central and eastern tropical Pacific Ocean. Over Indonesia, rainfall tends to increase while rainfall decreases over the central and eastern tropical Pacific Ocean. The normal easterly winds along the equator become even stronger. In general, the cooler the ocean temperature anomalies, the stronger the La Niña (and vice-versa).
 - Neutral:** Neither El Niño or La Niña. Often tropical Pacific SSTs are generally close to average. However, there are some instances when the ocean can look like it is in an El Niño or La Niña state, but the atmosphere is not playing along (or vice versa).
- During ENSO neutral conditions, surface trade winds blow **westward** across the equatorial Pacific Ocean. Blowing against the ocean's surface, these winds result in a westward current. **Hence, statement 2 is NOT correct.**

77. (a)

Explanation:

- Recently, the Ramgarh Vishdhar Wildlife Sanctuary received a nod from the National Tiger Conservation

Authority's (NTCA) technical committee to become the 4th Tiger Reserve of Rajasthan. This will be the **52nd Tiger Reserve of India. Hence, statement 1 is correct and 2 is NOT correct.**

- It was notified in the **Year 1982** and is spread over an area of 252.79 sq. Km. This Sanctuary is located in District Bundi in Eastern Rajasthan. **Hence, statement 3 is NOT correct.**
- The total area of 1,017 sq. km has been identified as the reserve area comprising two forest blocks of Bhilwara, territorial forest block of Bundi and Indargarh, which falls under the buffer zone of Ranthambore Tiger Reserve (RTR).
- Rajasthan has three tiger reserves with over 90 tigers at **Ranthambore Tiger Reserve (RTR)** in Sawai Madhopur, **Sariska Tiger Reserve (STR)** in Alwar, and **Mukundra Hills Tiger Reserve (MHTR)** in Kota.

78. (d)

Explanation:

- Recently, under the Pygmy Hog Conservation Programme (PHCP) eight captive-bred Pygmy Hogs, **the world's rarest and smallest wild pigs**, were released in the Manas National Park of Assam. **Hence, statement 1 is correct.**
- It is one of the very few mammals that build its own home, or nest, complete with a 'roof'. It is also **an indicator species**. Its presence reflects the health of its primary habitat, the tall, **wet grasslands of the region**. It thrives in wet grassland. **Hence, statements 2 and 3 are correct.**
- Habitat (grassland) loss and degradation, and illegal hunting are the major threats to this animal.
- About Pygmy Hog:**
 - Scientific Name: Porcula Salvania**

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○ Features:

- ◆ It is one of the very few mammals that build its own home, or nest, complete with a 'roof'.

○ Habitat:

- ◆ Once found along a narrow strip of tall and wet grassland plains on the Himalayan foothills – from Uttar Pradesh to Assam, through Nepal's terai areas and Bengal's duars. Today found mainly in a small population in Assam.

○ Protection Status:

- ◆ IUCN Red List: Endangered
- ◆ CITES: Appendix I
- ◆ Wildlife (Protection) Act, 1972: Schedule I

○ Threats:

- ◆ Habitat (grassland) loss and degradation, and illegal hunting.

79. (a)

Explanation:

- Absorbed carbon dioxide reacts with seawater and changes the pH of the ocean. This process is known as ocean acidification. **Hence, statement 1 is correct.**
- The rising acidity of the oceans threatens coral reefs by making it harder for corals to build their skeletons. **Hence, statement 2 is NOT correct.**

80. (b)

Explanation:

- Tsunamis have long-wavelength over deep waters and very low frequency. They have high frequency near the shore. **Hence, statement 1 is NOT correct.**
- The rate at which a tsunami loses its energy is inversely proportional to the wavelength of the waves. Since tsunami waves can have huge wavelengths, the loss of energy is very small. **Hence, statement 2 is correct.**
- Most of the tsunamis originate in the Pacific Ring of Fire, but the reason is not volcanic eruptions but frequent earthquakes. **Hence, statement 3 is NOT correct.**

81. (d)

Explanation:

Antarctic Treaty:

- The Antarctic Treaty was signed between 12 countries in Washington on 1st December 1959 for making the Antarctic Continent a demilitarized zone to be preserved for scientific research only. **Hence, statement 3 is correct.**

- The twelve original signatories are Argentina, Australia, Belgium, Chile, France, Japan, New Zealand, Norway, South Africa, the Union of Soviet Socialist Republics, the UK, and the US.
- It entered into force in 1961 and has since been acceded by many other nations. **Hence, statement 1 is NOT correct.**
- Currently, it has 54 parties. India became a member of this treaty in 1983. **Hence, statement 2 is correct.**
- Recently, the 60th anniversary of the Antarctic Treaty was celebrated.
- The purpose of the Antarctic Treaty System is to ensure in the interests of all mankind that Antarctica shall continue forever to be used exclusively for peaceful purposes and shall not become the scene or object of international discord.
- These agreements are legally binding and purpose-built for the unique geographical, environmental, and political characteristics of the Antarctic and form a robust international governance framework for the region.

82. (b)

Explanation:

Seafloor Spreading:

- Seafloor spreading is a geologic process in which tectonic plates—large slabs of Earth's lithosphere—split apart from each other.
- **Seafloor spreading occurs at divergent plate boundaries.** As tectonic plates slowly move away from each other, heat from the mantle's convection currents makes the crust more plastic and less dense. The less-dense material rises, often forming a mountain or elevated area of the seafloor. **Hence, statement 1 is NOT correct.**
- Seafloor spreading and other tectonic activity processes are the result of mantle convection. Mantle convection is the slow, churning motion of Earth's mantle. Convection currents carry heat from the lower mantle and core to the lithosphere. Convection currents also "recycle" lithospheric materials back to the mantle.

Mid-Ocean Ridges:

- Seafloor spreading occurs along mid-ocean ridges—large mountain ranges rising from the ocean floor.
- The **Mid-Atlantic Ridge**, for instance, separates the North American plate from the Eurasian plate, and the South American plate from the African plate.



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- The **East Pacific Rise** is a mid-ocean ridge that runs through the eastern Pacific Ocean and separates the Pacific plate from the North American plate, the Cocos plate, the Nazca plate, and the Antarctic plate.
- The **Southeast Indian Ridge** marks where the southern Indo-Australian plate forms a divergent boundary with the Antarctic plate.
- **Seafloor spreading is not consistent at all mid-ocean ridges.** Slowly spreading ridges are the sites of tall, narrow underwater cliffs and mountains. Rapidly spreading ridges have much more gentle slopes.
- **The Mid-Atlantic Ridge, for instance, is a slow spreading center.** It spreads 2-5 centimeters (.8-2 inches) every year and forms an ocean trench about the size of the Grand Canyon.
- The East Pacific Rise, on the other hand, is a fast spreading center. It spreads about 6-16 centimeters (3-6 inches) every year. There is not an ocean trench at the East Pacific Rise, because the seafloor spreading is too rapid for one to develop!
- The newest, thinnest crust on Earth is located near the center of mid-ocean ridge—the actual site of seafloor spreading. **The age, density, and thickness of oceanic crust increases with distance from the mid-ocean ridge.** Hence, statement 2 is correct.

Sea floor Spreading and Subduction:

- Seafloor spreading is just one part of plate tectonics. Subduction is another.
- Subduction happens where tectonic plates crash into each other instead of spreading apart.
- At subduction zones, the edge of the denser plate subducts, or slides, beneath the less-dense one. The denser lithospheric material then melts back into the Earth's mantle.
- Seafloor spreading creates a new crust. Subduction destroys old crust.
- The two forces roughly balance each other, so the shape and diameter of the Earth remain constant.

83. (a)

Explanation:

The following is the list of African mountains arranged from north to south:

- The **Atlas Mountains** are a mountain range in the Maghreb on the north west coast of Africa. It stretches around 2,500 km (1,600 mi) through Morocco, Algeria and Tunisia.

- Mount Cameroon** is one of Africa's largest volcanoes, rising to 4,040 metres (13,255 ft) above the coast of west Cameroon (Western Africa). It rises from the coast through tropical rainforest to a bare summit which is cold, windy, and occasionally dusted with snow. The massive steep-sided volcano of dominantly basaltic-to-trachybasaltic composition forms a volcanic horst.
- The **Drakensberg** is the name given to the eastern portion of the Great Escarpment, which encloses the central Southern African plateau. The Great Escarpment reaches its greatest elevation in this region – 2,000 to 3,482 metres (6,562 to 11,424 feet). It is located in South Africa. The name by which it is now known is Drakens Mountain, or Mountain of the Dragons.

Hence, option (a) is correct.



84. (b)

Explanation:

- The Ministry of Culture (MoC) and Ministry of Ports, Shipping and Waterways (MoPSW) signed a Memorandum of Understanding (MoU) for 'Cooperation in Development of National Maritime Heritage Complex (NMHC)' at Lothal, Gujarat. **Hence statement 3 is NOT correct.**



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- The idea is to create an edutainment (education with entertainment) approach for this destination that would be of great interest to the visitors. It would be developed as an international tourist destination, where the maritime heritage of India from ancient to modern times would be showcased. **Hence, statements 1 and 2 are correct.**
- The unique feature of NMHC is the recreation of ancient Lothal city, which is one of the prominent cities of the ancient Indus Valley Civilization.
- There will also be many pavilions where all coastal states in India and union territories can showcase their artefacts and maritime heritage.

85. (c)**Explanation:****Polar Vortex:**

- The polar vortex is a large area of low pressure and cold air surrounding both of the Earth's poles. It was also known as the Polar Pig. **Hence, statement 1 is correct.**
- The term "vortex" refers to the counterclockwise flow of air that helps keep the colder air near the Poles. It always exists near the poles, but weakens in summer and strengthens in winter. **Hence, statement 2 is correct.**

86. (c)**Explanation:****Dead Zones**

- Dead Zones also known as Hypoxic zone are areas of the sea where the lack of oxygen makes it difficult for organisms to survive. **Hence, statement 1 is correct.**
- There are many physical, chemical, and biological factors that combine to create dead zones, but nutrient pollution is the primary cause of those zones created by humans. **Hence, statement 2 is correct.**

87. (d)**Explanation:**

- Recently, the world's first Genetically Modified (GM) rubber plant developed by Rubber Research Institute of India was planted in Assam. **Hence statement 1 is NOT correct.**
- The GM rubber has additional copies of the gene MnSOD, or manganese-containing superoxide dismutase, inserted in the plant, which is expected to tide over the severe cold conditions during winter in the northeast.

The MnSOD gene has the ability to protect plants from the adverse effects of severe environmental stresses such as cold and drought. **Hence statement 2 is correct.**

- Natural rubber is a native of warm humid Amazon forests and is not naturally suited for the colder conditions in Northeast India, which is one of the largest producers of rubber in India. The growth of young rubber plants remains suspended during the winter months, which are also characterized by progressive drying of the soil. This is the reason for the long immaturity period of this crop in the region. **Hence statement 3 is correct.**

The Rubber Research Institute of India (RRII):

RRII was established in 1955. The Institute has attained a prestigious position in the international rubber scenario through its research contributions. RRII is a member of the International Rubber Research and Development Board (IRRDB) and actively participates in many international research programmes. RRII has played a significant role in India achieving high productivity.

Location:

The headquarters of RRII is located on a hillock, 8 km east of Kottayam town in Kerala State with a research farm attached to it.

88. (c)**Explanation:**

Wind: Wind is the movement of air caused by the uneven heating of the Earth by the sun.

- Differences in atmospheric pressure generate winds.
- At the Equator, the sun warms the water and land more than it does the rest of the globe.
- Warm equatorial air rises higher into the atmosphere and migrates toward the poles. **This is a low-pressure system.**
- At the same time, cooler, denser air moves over Earth's surface toward the Equator to replace the heated air. This is a high-pressure system.
- Winds generally blow from high-pressure areas to low-pressure areas.

Wind Zones: The Earth contains five major wind zones: **polar easterlies, westerlies, horse latitudes, trade winds, and the doldrums.**

- **Horse Latitudes**
 - The horse latitudes are a narrow zone of warm, dry climates between westerlies and the trade winds.

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- Horse latitudes are about 30 and 35 degrees north and south. **Many deserts, from the rainless Atacama of South America to the arid Kalahari of Africa, are part of the horse latitudes. Hence, statement 1 is correct.**
- The prevailing winds at the horse latitudes vary, but are usually light. Even strong winds are often short in duration.

Layers of the Atmosphere

- Earth's atmosphere has a layered structure.
- From the ground toward the sky, the layers are the **troposphere, stratosphere, mesosphere, thermosphere, and exosphere.**
- Another layer, called the ionosphere, extends from the mesosphere to the exosphere. Beyond the exosphere is outer space. The boundaries between atmospheric layers are not clearly defined, and change depending on latitude and season.

Troposphere

- The troposphere is the lowest atmospheric layer. On average, the troposphere extends from the ground to about 10 kilometers (6 miles) high, ranging from about 6 kilometers (4 miles) at the poles to more than 16 kilometers (10 miles) at the Equator.
- **The top of the troposphere is higher in summer than in winter. Hence, statement 2 is correct.**
- Almost all weather develops in the troposphere because it contains almost all of the atmosphere's water vapor.
- Fast-moving, high-altitude winds called jet streams blow around the planet near the upper boundary of the troposphere. Jet streams are extremely important to the airline industry. Aircraft save time and money by flying in jet streams instead of the lower troposphere, where air is thicker.

89. (b)

Explanation:

Temperature of Ocean Waters

- Ocean waters get heated up by solar energy just as land.
- The process of heating and cooling of the oceanic water is slower than land.

Factors Affecting Temperature Distribution

The factors which affect the distribution of temperature of ocean water are:

- (i) **Latitude:** the temperature of surface water decreases from the equator towards the poles because the amount of insolation decreases poleward.
- (ii) **Unequal distribution of land and water:** the oceans in the northern hemisphere receive more heat due to their contact with larger extent of land than the oceans in the southern hemisphere.
- (iii) **Prevailing wind:** the winds blowing from the land towards the oceans drive warm surface water away from the coast resulting in the upwelling of cold water from below.
It results in longitudinal variation in the temperature. Contrary to this, the onshore winds pile up warm water near the coast and this raises the temperature.
- (iv) **Ocean currents:** warm ocean currents raise the temperature in cold areas while the cold currents decrease the temperature in warm ocean areas.

Horizontal and Vertical Distribution of Temperature

- The temperature-depth profile for the ocean water shows how the temperature decreases with the increasing depth.
- There is a boundary region between the surface waters of the ocean and the deeper layers. This boundary region, where a rapid decrease in temperature takes place, is called the thermocline.
- The temperature structure of oceans over middle and low latitudes can be described as a three-layer system.
- The first layer represents the top layer of warm oceanic water and it is about 500m thick with temperatures ranging between 20° and 25° C.
- This layer, within the tropical region, is present throughout the year but in mid latitudes develops only during summer.
- The second layer called the thermocline layer lies below the first layer and is characterised by rapid decrease in temperature with increasing depth. **Hence, statement 2 is correct.**
- The thermocline is 500 -1,000 m thick.
- The third layer is very cold and extends upto the deep ocean floor.
- In the Arctic and Antarctic circles, the surface water temperatures are close to 0° C and so the temperature change with the depth is very slight. Here, only one layer of cold water exists, which extends from surface to deep ocean floor. **Hence, statement 3 is correct.**



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- The average temperature of surface water of the oceans is about 27°C and it gradually decreases from the equator towards the poles.
- The rate of decrease of temperature with increasing latitude is generally 0.5°C per latitude.
- The oceans in the northern hemisphere record relatively higher temperature than in the southern hemisphere.
- The highest temperature is not recorded at the equator but slightly towards north of it. **Hence, statement 1 is NOT correct.**
- The average annual temperatures for the northern and southern hemisphere are around 19° C and 16° C respectively. This variation is due to the unequal distribution of land and water in the northern and southern hemispheres.

90. (c)

Explanation:

Integrated Power Development Scheme:

- IPDS was launched in **2014**. **Power Finance Corporation Ltd. (PFC)** is the nodal agency for the implementation of this scheme. Strengthening of sub-transmission and distribution networks in the **urban areas**, metering of distribution transformers/feeders/consumers in the urban areas, and **IT enablement of the distribution sector** are the focused areas under this scheme. Underground cabling to include additional demand of States and smart metering solution for performing UDAY States and Solar panels on Govt. buildings with net-metering are also permissible under the scheme. **Hence statements 1 and 2 are correct.**
- **All Power Distribution Companies (Discoms)** are eligible for financial assistance under the scheme. **Hence statement 3 is NOT correct.**

91. (b)

Explanation:

- Recently, **Shyam Sundar Jyani**, a Rajasthan-based climate activist won the prestigious **United Nations' Land for Life Award** for his environment conservation concept, Familial Forestry.
- Familial Forestry means transferring the care of the tree and environment in the family so that a tree becomes a part of the family's consciousness.
- Every two years, they organise the Land for Life Award. The Award recognizes excellence and innovation in efforts towards land in balance.

- The theme for the Land for Life award, 2021 is 'Healthy Land Healthy Life'. **Hence, option (b) is correct.**

United Nations Convention to Combat Desertification (UNCCD):

It was established in 1994, the sole legally binding international agreement linking environment and development to sustainable land management.

- The Delhi Declaration of 2019, signed by 14th CoP of the UNCCD, called for better access and stewardship over land, and emphasised gender-sensitive transformative projects.

92. (c)

Explanation:

Coral Reef

- Red Sea coral reefs are the northernmost in the Indian Ocean. About 300 hard coral species have been recorded from the Red Sea as a whole. **Hence, statement 1 is NOT correct.**
- Extreme surface temperature, brought by El Nino, is endangering the Great Barrier Reef. El Nino causes anomalously cold or hot ocean temperatures around the globe which can threaten the diverse nature of Great Barrier Reef. **Hence, statement 2 is correct.**
- Coral reefs are found throughout the oceans, from deep, cold waters to shallow, tropical waters. Based on current estimates, shallow water coral reefs occupy somewhere between 284,000 and 512,000 km² of the planet which is less than the area occupied by cold-water (deep) coral reefs. **Hence, statement 3 is correct.**

93. (c)

Explanation:

- Recently, a Memorandum of Understanding (MoU) was signed between the Gujarat Maritime University and International Financial Services Centres Authority in GIFT (Gujarat International Finance Tec-City) City. The idea is to create a world-class arbitration center focused on maritime and shipping disputes that can help resolve commercial and financial conflicts between entities having operations in India. **Hence statement 1 is correct.**
- This will be the first center of its kind in the country that will manage arbitration and mediation proceedings with disputes related to the maritime and shipping sector. There are over 35 arbitral institutions in India, however,



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none focus exclusively on the disputes related to the maritime sector. The arbitration involving Indian players is now heard at the Singapore Arbitration Centre. **Hence statement 2 is correct.**

94. (c)

Explanation:

Dew:

- When the moisture is deposited in the form of water droplets on cooler surfaces of solid objects (rather than nuclei in air above the surface) such as stones, grass blades and plant leaves, it is known as dew.
- The ideal conditions for its formation are **clear sky, calm air, high relative humidity, and cold and long nights. Hence, statement 1 is correct.**
- For the formation of dew, it is necessary that the dew point is above the freezing point. **Hence, statement 2 is correct.**

Frost:

- Frost forms on cold surfaces when condensation takes place below freezing point (0°C), i.e. the dew point is at or below the freezing point.
- The excess moisture is deposited in the form of minute ice crystals instead of water droplets.
- The ideal conditions for the formation of white frost are the same as those for the formation of dew, except that the air temperature must be at or below the freezing point.

95. (c)

Explanation:

Relief of the Ocean Floor

- **Continental Shelf**
 - The continental shelf is the extended margin of each continent occupied by relatively shallow seas and gulfs.
 - It is the **shallowest part of the ocean** showing an average gradient of 1° or even less.
 - The shelf typically **ends at a very steep slope**, called the shelf break.
 - The width of the continental shelves vary from one ocean to another. The **average width of continental shelves is about 80 km.**
 - The shelves are almost absent or very narrow along some of the margins like the coasts of Chile, the

west coast of Sumatra, etc. On the contrary, the **Siberian shelf in the Arctic Ocean**, the largest in the world, stretches to **1,500 km in width.**

- The depth of the shelves also varies. It may be as shallow as 30 m in some areas while in some areas it is as deep as 600 m.
- The continental shelves are covered with variable thicknesses of sediments brought down by rivers, glaciers, wind, from the land and distributed by waves and currents.
- **Massive sedimentary deposits received** over a long time by the continental shelves, become the **source of fossil fuels. Hence, statement 1 is correct.**

Seamount

- It is a mountain with pointed summits, rising from the seafloor that does not reach the surface of the ocean. **Hence, statement 2 is correct.**
- Seamounts are volcanic in origin.
- These can be 3,000-4,500 m tall. The Emperor seamount, an extension of the Hawaiian Islands in the Pacific Ocean, is a good example.

96. (a)

Explanation:

Minimum Support Price (MSP)

The MSP is the rate at which the government purchases crops from farmers, and is based on a calculation of at least one-and-a-half times the cost of production incurred by the farmers.

Crops under MSP:

- The Commission for Agricultural Costs & Prices (CACP) recommends MSPs for 22 mandated crops and fair and remunerative price (FRP) for sugarcane.
- The mandated crops include 14 crops of the Kharif season, 6 rabi crops and 2 other commercial crops.
- In addition, the MSPs of toria and de-husked coconut are fixed on the basis of the MSPs of rapeseed/mustard and copra, respectively.
- **The list of crops is as follows:**
 - **Cereals (7):** Paddy, wheat, barley, jowar, bajra, maize and ragi,
 - **Pulses (5):** Gram, arhar/tur, moong, urad and lentil,
 - **Oilseeds (8):** Groundnut, rapeseed/mustard, toria, soybean, sunflower seed, sesamum, safflower seed and niger seed,



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- **Commercial Crops:** Raw cotton, Raw jute, Copra, De-husked coconut. **Hence, statement 2 is NOT correct.**
- Sugarcane (Fair and remunerative price).

Factors for Recommending the MSP:

- **While recommending the price the CACP analyzes:**
 - demand and supply;
 - cost of production;
 - price trends in the market, both domestic and international; **Hence, statement 1 is correct.**
 - inter-crop price parity;
 - terms of trade between agriculture and non-agriculture;
 - a minimum of 50 per cent as the margin over the cost of production; and
 - likely implications of MSP on consumers of that product.
- **Three Kinds of Production Cost:**
 - The CACP projects three kinds of production costs for every crop, both at the state and all-India average levels.
 - ◆ A2':
 - Covers all paid-out costs directly incurred by the farmer in cash and kind on seeds, fertilisers, pesticides, hired labour, leased-in land, fuel, irrigation, etc.
 - ◆ 'A2+FL'
 - Includes A2 plus an imputed value of unpaid family labour.
 - ◆ 'C2'
 - It is a more comprehensive cost that factors in rentals and interest forgone on owned land and fixed capital assets, on top of A2+FL.
 - CACP considers both A2+FL and C2 costs while recommending MSP.
 - CACP reckons only A2+FL cost for return.
 - However, C2 costs are used by CACP primarily as benchmark reference costs (opportunity costs) to see if the MSPs recommended by them at least cover these costs in some of the major producing States.
 - The Cabinet Committee on Economic Affairs (CCEA) of the Union government takes a final

decision on the level of MSPs and other recommendations made by the CACP.

Commission for Agricultural Costs and Prices (CACP)

- The Commission for Agricultural Costs and Prices (CACP) is an attached office of the Ministry of Agriculture and Farmers Welfare, Government of India. It came into existence in January 1965.
- It is an advisory body whose recommendations are not binding on the Government. **Hence, statement 3 is NOT correct.**

97. (c)

Explanation:

- Surface oceanic currents are sometimes wind driven and develop their typical clockwise spirals in the northern hemisphere and counter-clockwise rotation in the southern hemisphere due to imposed wind stresses.
- **Surface currents make up only 8% of all water in the ocean.** They are generally restricted to the upper 400m (1,300 ft) of ocean water, and are separated from lower regions by varying temperatures and salinity which affect the density of the water, which in turn, defines each oceanic region. **Hence, statement 3 is NOT correct.**
- Because the movement of deep water in ocean basins is caused by density driven forces and gravity, deep waters sink into deep ocean basins at high latitudes where the temperatures are cold enough to cause the density to increase. **Hence, statement 1 is correct.**
- **Thermohaline circulation drives deep ocean circulation.** Thermo means heat and haline refers to salinity. Differences in temperature and in salinity change the density of seawater. So thermohaline circulation is the result of density differences in water masses because of their different temperature and salinity.
- **Downwelling:** Changes in temperature and salinity of seawater take place at the surface. Water becomes dense near the poles. Cold polar air cools the water and lowers its temperature, increasing its salinity. Fresh water freezes out of seawater to become sea ice, which also increases the salinity of the remaining water. This very cold, very saline water is very dense and sinks. This sinking is called downwelling.



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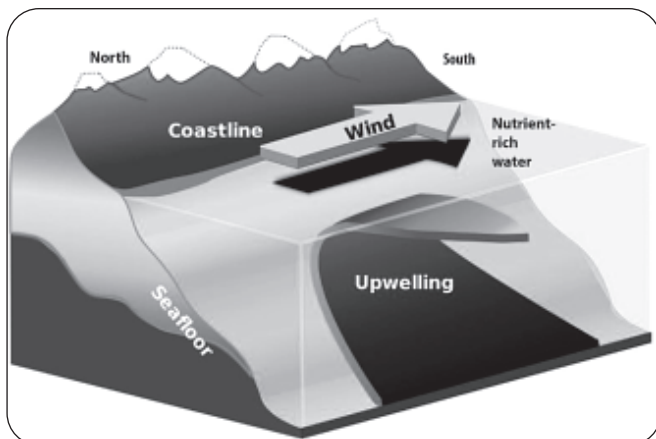
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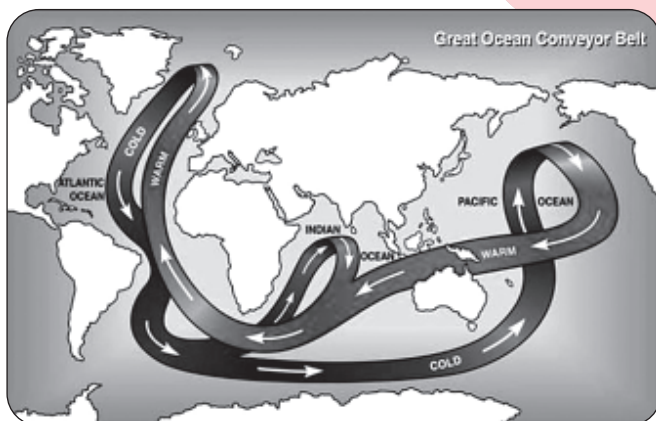
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- **Upwelling:** Upwelling occurs along the coast when wind blows water strongly away from the shore. This leaves a void that is filled by deep water that rises to the surface. Upwelling is extremely important where it occurs. During its time on the bottom, the cold deep water has collected nutrients that have fallen down through the water column. Upwelling brings those nutrients to the surface. Those nutrients support the growth of plankton and form the base of a rich ecosystem. California, South America, South Africa, and the Arabian Sea all benefit from offshore upwelling. **Hence, statement 2 is correct.**



98. (d)

Explanation:

Giant kelp forests are among Earth's most productive habitats, and their great diversity of plant and animal species supports many fisheries around the world. They are the world's largest marine plants and regularly grow up to 35 meters (115 feet) tall; the largest giant kelp on record stood 65 meters (215 feet) tall.

Distribution:

- Worldwide in temperate to polar latitudes. **Hence, statement 1 is NOT correct.**
- Kelp Forests are **underwater ecosystems** formed in **shallow water** by the dense growth of several different species known as kelps. Though they look very much like plants, kelps are actually extremely **large brown algae**. **Hence, statement 2 is NOT correct.**
- Climate change is altering marine habitats such as kelp forests.
- Genetic evidence reveals that most kelps **reinvaded the Arctic** from the Atlantic Ocean quite recently (approximately 8,000 years ago, following the last Ice Age).
- As a result, most kelps in the Arctic are living in waters colder than their optimal temperature. **Ocean warming will** also move conditions closer to temperatures of maximum growth, and could increase the productivity of these habitats.
- As waters warm and sea ice retreats, more light will reach the seafloor, which will benefit marine plants. Researchers predict a **northern shift of kelp forests as ice retreats**.
- Underwater Arctic forests are expanding due to global warming. **Hence, statement 3 is NOT correct.**

99. (d)

Explanation:**New Atlantic Charter (2021)**

- The new Atlantic Charter is a 600-word document that reaffirms the joint commitment of the US and the UK on a range of issues. **Hence, statement 1 is NOT correct.**
- It is a statement of principles, a promise that the UK and the US would meet the challenges of their age together. It calls for both countries to adhere to the rules-based international order.
- The new charter focuses on **climate change and the need to protect biodiversity**. With references to emerging technologies, cyberspace and sustainable global development. **Hence, statement 2 is NOT correct.**
- It calls on Western allies to oppose interference through disinformation or other malign influences, including in elections.

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- It ranks the threats to democratic nations in a technological era.
- It vows that as long as there are nuclear weapons, North Atlantic Treaty Organisation (NATO) will remain a nuclear alliance. It is not a successor pact to NATO. Hence, statement 3 is NOT correct.

Atlantic Charter (1941):

- The Atlantic Charter was a joint declaration released by US President Franklin D. Roosevelt and British Prime Minister Winston Churchill on 14th August 1941 (during World War II) following a meeting of the two heads of government in Newfoundland.
- The Atlantic Charter was subsequently incorporated by reference in the Declaration of the United Nations in 1942.
- The Atlantic Charter provided a broad statement of US and British war aims such as:
 - They desired no territorial changes without the free assent of the people concerned.
 - They respected every people's right to choose their own form of government and wanted sovereign rights and self-government restored to those forcibly deprived of them.
 - They would try to promote equal access for all states to trade and to raw materials.
 - They hoped to promote worldwide collaboration so as to improve labour standards, economic progress, and social security.
 - With the destruction of "Nazi tyranny," (Germany) they would look for a peace under which all nations could live safely within their boundaries, without fear or want.
 - Under such peace, the seas should be free.
 - Pending general security through renunciation of force, potential aggressors must be disarmed.

Collective Security

- Collective security, a system by which states have attempted to prevent or stop wars.
- Under a collective security arrangement, an aggressor against any one state is considered an aggressor against all other states, which act together to repel the aggressor.

100. (c)

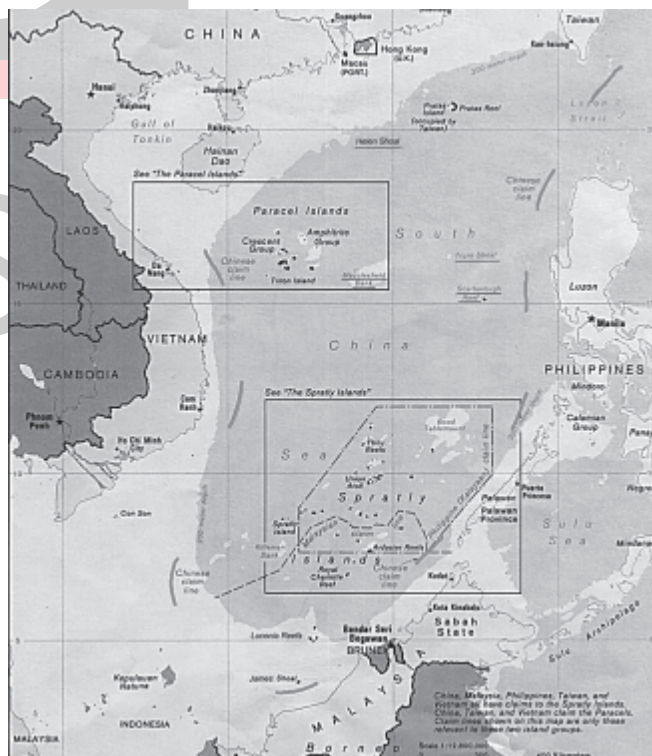
Explanation:

Spratly Islands dispute

- The ongoing territorial dispute is between **China, Taiwan, Vietnam, the Philippines, and Malaysia** concerning the ownership of the Spratly Islands archipelago and nearby geographical features like corals reefs, cays etc. Hence, statement 2 is NOT correct.
- Although Spratly islands are largely uninhabited, the islands may have large reserves of untapped natural resources including oil. Hence, statement 1 is correct.

Paracel Islands

- Located in the South China Sea, almost equidistant from China and Vietnam.
- Beijing says that references to the Paracel Islands as a part of China sovereign territory can be found in 14th century writings from the Song Dynasty.
- Vietnam on the other hand, says that historical texts from at least the 15th century show that the islands were a part of its territory.
- With increased tensions accelerated by Colonial powers, China and Vietnam fought over their territorial disputes in January 1974 after which China took over control of the islands.



Hence, statement 3 is correct.



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